Exclusion by design: the effects of limiting neighbourhood access on incumbent residents*

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Abstract

Neighbourhoods shape economic and life outcomes. Existing research relies on selective residential moves to identify neighbourhood effects. We study these effects using a plausibly exogenous change in composition induced by the *Rotterdam Wet*, a policy barring workless individuals with less than six years of residency in the region from moving into selected disadvantaged neighbourhoods. The policy also displaced targeted areas' residents meeting the work and tenure criteria at lease renewal. In targeted neighbourhoods, the policy led to a persistent increase in the share of employed residents and a decrease in welfare recipiency and criminal involvement, driven by changes in the type of entrants. We observe a behavioural response among incumbent residents, who increase their employment, decrease their welfare participation and criminal behaviour. Exploiting the sharp discontinuity in policy applicability at the six-year residency threshold, we show that incumbent residents react to the displacement threat by exiting welfare and informal work to take up formal employment. Future analyses exploiting street-level data will establish the role of local peer composition on labour outcomes and criminal offences.

JEL Classification: J18, J62, J61.
Keywords: neighbourhood effects, place-based policy, housing allocation, labour supply, welfare participation, crime.

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1 Introduction

The neighbourhood we live in shapes our economic opportunities, social networks, and overall well-being. A large body of evidence shows that relocating low-income families to better neighbourhoods substantially improves long-run outcomes for children (Chetty et al., 2016). While this literature has focused on the effects of moving into new environments, less is known about the reverse dynamic: how changes in neighbourhood composition due to inflows of new residents affect the outcomes and behaviour of incumbent residents. This question has far-reaching implications for the provision of place-based and welfare policies. Neighbourhood composition is known to shape peer group formation (Åslund et al., 2011) and may trigger moving responses by residents (Card et al., 2008). The magnitude and direction of these effects are decisive factors for policy effectiveness. For example, reforms in housing allocation or refugee settlement programs can significantly shift the layout of a neighbourhood in a short period, with potentially unintended consequences for those already living there.

We exploit the *Rotterdam Wet*, a Dutch policy implemented in 2006 impeding workless individuals having lived less than six years in the region of Rotterdam from moving into a selection of disadvantaged neighbourhoods. These restrictions also prevented current residents meeting the criteria from renewing their leases. The policy applies to the regulated rental sector, representing the vast majority of housing in targeted areas. Using rich administrative data, we show that the policy restrictions induced a persistent decline in welfare recipiency and an increase in employment in treated neighbourhoods, driven primarily by changes in the inflow composition of residents. Similar neighbourhoods in the same city were unaffected, although we identify areas into which the policy likely spilled over. Rather than relying on selective residential moves, our empirical setting leverages a plausibly exogenous change in composition and incentives caused by restrictions in residents' moving options. Leveraging the six-year settlement tenure cut-off, we can disentangle how incumbent residents respond - economically and behaviourally - to the policy-induced incentives, and the associated changes in the socioeconomic layout of their neighbourhood.

Identification of the medium-term effects for incumbent residents relies on event-study difference-in-differences (DiD) comparing adult individuals in treated and control neighbourhoods in the region surrounding Rotterdam. We show strong pre-trend balance and address concerns about spillovers and other concurrent neighbourhood interventions. To pin down the threat effects of the the policy, we use a triple-difference comparing adults in treated and control areas straddling the 6 year tenure threshold. Future analyses will exploit street-level data to determine the role of local peers in the effects we estimate. In the spirit of Carrillo et al. (2023), we rely on within-neighbourhood variation in policy exposure due to children's age to measure the effects of the policy on youth outcomes.

In its immediate impact, we find that the *Rotterdam Wet* led to a persistent change in the composition in targeted neighbourhoods. The share of residents receiving welfare benefits decreased by 2.5 percentage points (12.9%), and the share in employment increased by 3 pp (5%) within 2 years of implementation. The fraction of residents having committed a crime decreased by 0.5 pp (20%) four years after treatment. The shifts in composition are primarily driven by changes in the type of entrants. The policy reduced the entry of individuals receiving welfare, and attracted low-income workers instead. We observe no changes in the ethnic composition of targeted neighbourhoods, and a slight increase in the share of single-head households.

Building on these findings, we focus our analysis on the effects of the policy on incumbent residents. In an intention-to-treat (ITT) framework, we estimate the effects of the policy on individuals living in the same neighbourhood in the two years before treatment, fixing their treatment status based on their place

of residence in those years. Long-term residents (incumbents) increase employment by 0.6 pp (1 %) and reduce welfare participation by 0.5 pp (2.6%) 2 years after treatment. We find no meaningful substitution from welfare benefits into other government benefits programs. Exploiting within-neighbourhood variation in the age of children when the policy was implemented, we estimate the effects on schooling outcomes for young children passing the *cito* test (at age 11-12), and teenagers finishing high school (the *eind examen*, at age 15-18). Consistent with the literature on long-run place effects, we find null and imprecisely estimated effects on test scores, the probability to take exams, and the probability to pass exams both for young and teenage children.

To strengthen our identification and investigate the mechanisms, we exploit the sharp cut-off in policy applicability at the 6 year tenure threshold to run a triple-difference (DDD) estimator. The DDD estimator alleviates concerns on unobserved confounders affecting our results, and reveals a strong behavioural response by individuals directly affected by the policy's restrictions. Directly affected individuals - living since less than 6 years in treated neighbourhoods - increase formal employment participation by 2.2 pp (3.3%), do not take up more benefits, and reduce informal or unpaid work by 2.1 pp (13.7%). We also find large but imprecisely estimated negative effects on criminal offences. Although these results suggest a large role for incentives in the employment effects we find, we cannot rule out that changes in local peer composition did not affect incumbent residents. Future analyses exploiting dwelling-level data will reveal whether this is the case.

We investigate treatment effects heterogeneity using causal forests (Athey et al., 2019). The employment effects are driven by women, and individuals between the age of 20 and 35. Symmetrically, younger women are more likely to exit non-employment or informal employment. For both outcomes, we find a light income gradient in the magnitude of the effects. These patterns are both consistent with a behavioural response to the policy, and neighbourhood peer effects.

Treatment may spill over to neighbouring areas, changing their composition. We identify spillover neighbourhoods as the areas with highest pre-policy migration flows to treated neighbourhoods. Re-running the analysis with residents of spillover neighbourhoods coded as treated shows weak negative effects on welfare participation and weak positive effects on employment. All spillover locations are situated in the direct surroundings of the treated area. These findings likely indicate that the policy affected long-term residents in adjacent locations. A placebo exercise using the triple difference specification displays null effects on all outcomes.

Our findings relate to a literature on welfare dependence, peer effects, and activation programmes. Card and Hyslop (2005) structurally estimate the effect of an employment earnings subsidy (the SSP) aimed at breaking welfare dependence. Eligibility for their subsidy elapses after 1 year if treated individuals do not find a job. The dynamic incentives of SSP echo those of our policy, incentivising work through housing restrictions. The effects of the *Rotterdam Wet* could also be caused by changes in the local peer composition. Dahl, Kostøl and Mogstad (2014) coin the term 'welfare cultures' to designate an *ensemble* of information, norms, and beliefs enticing the reliance on welfare benefits for subsistence.[1] Little is known, however, about how social composition outside of families affects labour market and welfare participation. We show that welfare cultures may have a local dimension, where employment and welfare decisions are influenced by neighbourhood-level compositional factors and incentives.

[1] Welfare cultures are related to peer effects, which are known to be drivers of individual outcomes and decisions. Prominent examples related to our study include: Carrell et al. (2018), Åslund et al. (2011), Case and Katz (1991), Damm and Dustmann (2014), and Dahl, Løken and Mogstad (2014).

A burgeoning literature shows that places causally affect youth's long-term outcomes.[2] School quality (Laliberté, 2021) and peers (Deutscher, 2020; Chetty et al., 2026) drive place effects on children. The evidence on place effects for adults is weaker, with some mild effects found on health (Ludwig et al., 2012, Hasager and Jørgensen, 2024) and employment (Billings et al., 2026). Our findings show that neighbourhood composition affects *adult* incumbent residents through their welfare, employment, and criminal behaviour, suggesting that contextual effects matter for adults beyond their mental health outcomes. Our findings lend support to the idea that local composition may fuel socio-economic disparities.[3]

Residential choices and local amenities are central in shaping local-level outcomes. Changes in neighbourhood composition may affect the quality of neighbourhoods through schools (Bayer et al., 2007; Neri, 2024; Blanco and Neri, 2023), crime levels (Autor et al., 2017; Aliprantis and Hartley, 2015), and the endogenous development of amenities (Diamond, 2016; Redding and Sturm, 2024). Local social composition can also be viewed as an amenity, with stark implications for sorting patterns (Card et al., 2008). The policy we analyse modifies expectations about local social composition in the future, which may influence households in their residential choices (Bayer et al., 2016), but may backfire through stigma effects (Govind et al., 2025; Koster and Van Ommeren, 2024). We show that selecting residential inflow successfully reduced crime rates and welfare benefits reliance, whilst indirectly affecting incumbent residents through behavioural responses. Our focus on long-term residents, as in Baum-Snow et al. (2019) and Böhlmark and Willén (2020), sheds light on the role of place effects and social composition on incumbents, which although they represent a large majority in the population, receive less attention than movers in recent research (Zabek, 2024).

The article is structured as follows. Sections 2 and 3 discuss the institutional context of Rotterdam, along with administrative data sources we use. Section 5 details the empirical strategies used to identify the effects of the policy on different demographic groups. Sections 6 present and discuss the results, before concluding in section 7.

2 Institutional setting

We study the effects of the *Wet Bijzondere Maatregelen Grootstedelijke Problematiek* or 'Act on Extraordinary Measures on Urban Problems', a law allowing local governments to limit the inflow of residents without employment income in selected disadvantaged neighbourhoods. The law, commonly referred to as the *Rotterdamwet*, was implemented in 2006 in Rotterdam before being rolled out to other Dutch cities.[4] The law can be applied in neighbourhoods with more than 5.000 housing units, in which 25% of residents receive government benefits or 45% of households have a low income. The competent minister must grant a permit to implement the law, which must be renewed every four years. The stated aim of the policy is to improve the liveability and safety of distressed urban localities. Policymakers argue that by reducing the inflow of jobless residents - and attracting employed individuals instead - the social mix and liveability in targeted neighbourhoods can be improved.

The Rotterdamwet (henceforth RW) has been subject to numerous changes since its introduction in 2005. In its final iteration, the law allows municipalities to change the inflow of residents in three ways. (i) Individuals who have lived in the region for less than six years must have some employment income to move

[2] See Chetty et al. (2018), Chyn and Katz (2021), Chyn (2018), Chyn et al. (2023), Deutscher (2020), Britto et al. (2022), Kawano et al. (2024), Webbink et al. (2025).

[3] For discussions on spatial disparities and their interaction with across-individual disparities see Gaubert et al. (2021), Autor (2020), Bartik (2020), Chetty et al. (2014), Diamond and Gaubert (2022), Diamond (2016), Beekers (2024).

[4] Namely Amsterdam, Zaandam, Utrecht, The Hague, and Tilburg.

into targeted neighbourhoods. (ii) Individuals with occupations in services, manufacturing, social work, or education may be granted priority for housing. (iii) A housing permit may be denied if authorities have a reason to think that it may lead to increased nuisance or crime. Provision (i) was consistently implemented between 2006 and 2022, whereas provision (iii) was implemented in certain streets from 2014 onwards. In 2022, the municipality discontinued the enforcement of the law in Rotterdam. For the purposes of this paper, we focus on provision (i).

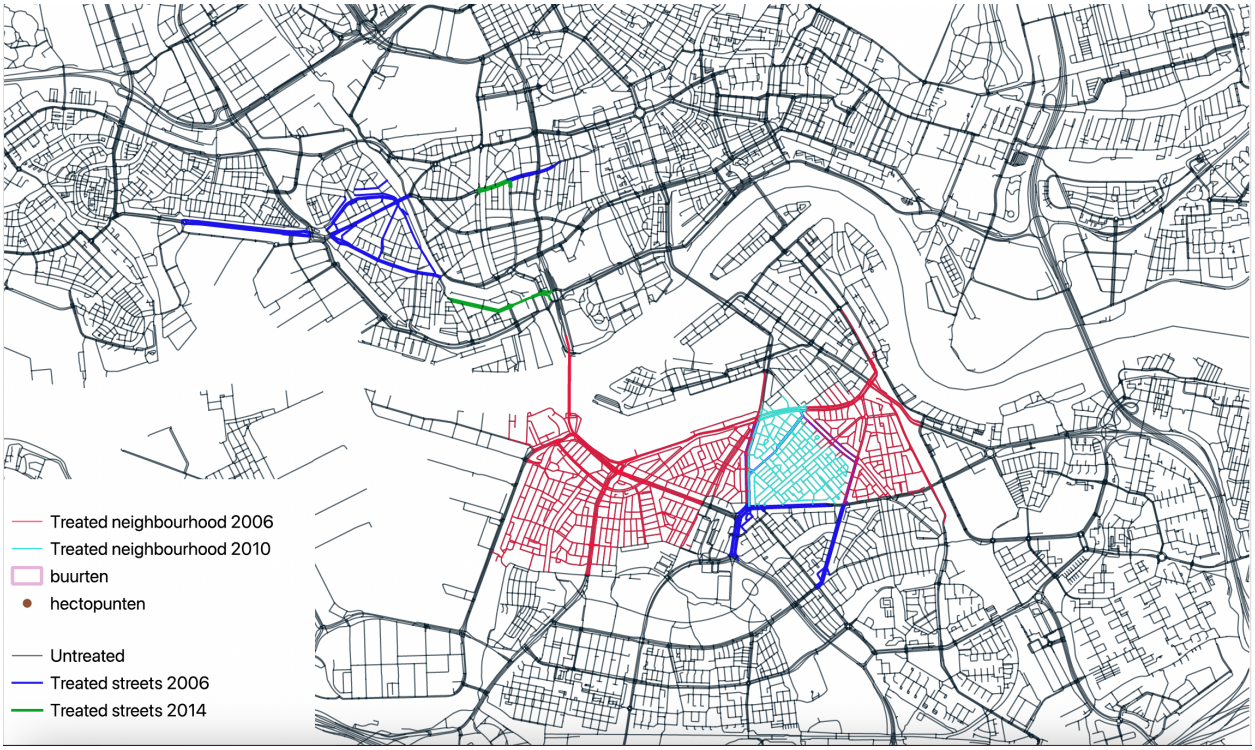
The RW applies to all individuals renting housing in the regulated sector (under €699 per month in 2014) in targeted areas. Regulated housing includes private and public housing units. The regulated sector accounts for 53.5% of all housing units in Rotterdam in 2022 (CBS, 2022a).[5] Figure 1 gives an overview of the location of fully (red and cyan) and partially treated (green and blue) neighbourhoods.[6] All housing units in the regulated sector in fully treated neighbourhoods are subjected to the policy.[7] In partially treated neighbourhoods, the policy applies to a selection of streets (see Table A1 for a complete overview). We exclude these neighbourhoods from our analysis. Five neighbourhoods (Oud-Charlois, Carnisse, Hillesluis, Tarwewijk, and Bloemhof) were fully treated, and two other neighbourhoods (Tussendijken and Spangen) were partially treated from 2006 until 2022. The policy was partially applied from 2014 until 2022 in three additional neighbourhoods (Middelland, Delfshaven, and Bospolder). Bloemhof was partially treated until 2010, but given that a majority of streets were treated between 2006 and 2010 we consider it as fully treated from 2006. Similarly, one neighbourhood (Nieuwe Westen) had part of a street being treated between 2014 and 2018. We consider this latter neighbourhood as untreated. The results are unaffected by this choice.

[5] This figure was likely much higher at the time of the implementation of the policy. In 2005 72% of housing in Rotterdam was dedicated to rental housing (CBS, 2005), of which an overwhelming majority falls under the regulated sector.

[6] See A for a map of treated areas matched with the spatial distribution of adult individuals on any type of government benefit in Rotterdam. See Section E for further details on the Dutch welfare system.

[7] In 2005, 72% of all housing in treated areas was dedicated to rental (CBS, 2005), of which the overwhelming majority falls under the regulated sector. In 2009, this number was 65%. 37,6% of housing in treated neighbourhoods was owned by housing corporations in 2009 (CBS, 2009).

Figure 1: **Treated areas in Rotterdam 2006-2018**



Note: **fully treated** neighbourhoods (as of 2006) are represented in **red**. Fully treated neighbourhoods (as of 2010) are represented **cyan**. Treated streets (as of 2006) are in **blue**. Treated streets (as of 2014) are in **green**.

In addition to the legal rules constraining the application of the law, municipalities have discretion over which neighbourhoods are targeted. From 2005 until around 2012, the municipality used a 'problem cumulation score', a combination of 12 objective indicators used to predict the liveability of neighbourhoods in Rotterdam (see Table A2 for list of criteria). Neighbourhoods with a higher score would have a higher chance of being treated. From 2012 onwards, the municipality switched to a 'liveability barometer' to select treated areas. The calculations underlying these numbers are unknown to us. Figure A1 shows that multiple neighbourhoods with similar shares of individuals on welfare as treated neighbourhoods were not selected for treatment. The same pattern holds when plotting on a map the 'liveability barometer' (Figure A2), and the 'neighbourhood signalling index' (Figure A4).

A number of other policies aimed at improving neighbourhood-level outcomes were implemented in fully treated neighbourhoods in the 2000s. The *Nationaal Programma Rotterdam Zuid* (NPRZ) was launched in 2013-2014. The policy earmarked funds to renovate social housing and offer additional school hours for children in the south of Rotterdam. The timing of the policy implies it does not directly affect our study. The *Vogelaarwijken* initiative was introduced in 2007 to invest in disadvantaged neighbourhoods throughout the Netherlands. Similarly, redlining in the late 90s and large housing renovations affected Rotterdam. Our understanding is that these policies affected neighbourhoods both in the control and in the treated group of neighbourhoods, mitigating concerns that we may be picking up the effect of other policies in our analyses. The DDD specification's results should dispel any concerns regarding these confounders.

3 Data

Our study is based on restricted-access microdata from CBS (*Centraal Bureau voor de Statistiek)*, covering the universe of registered residents in the Netherlands. We construct a panel of individuals having ever lived in Rotterdam, its metropolitan region, and the Hague between 1999 and 2019 (2.6 m individuals). Using an encrypted identifier from the municipal registers (*GBA-Gemeentelijke Basisregistratie*), we can link individuals to a rich set of administrative data sources. For our analyses, we restrict the time frame to 2002-2015.

Housing We match every individual in the panel to a housing unit, from which we construct the housing spells. Spells are observed for all individuals registered at a municipality. This is mandatory for all working individuals, and for those wishing to stay in the Netherlands for more than six months. The opening of a bank account, as well as access to virtually all government services is conditional on registering at the municipality. The strong incentives to register and be part of the municipality registry ensure good quality data.[8]

Demographics For each individual, we obtain demographic characteristics on age, gender, migration background, and country of origin. We can link parents to children, and map partnerships, observe the number of children per household, as well as each individual's position in the household.[9] We define individuals as first generation migrants if they are born outside of the Netherlands and at least one of their parents is born abroad. Second generation migrants are born in the Netherlands, but have at least one parent born abroad. Individuals' origin is defined as the place of birth for first generation migrants. For second generation migrants, the place of birth of the mother determines the origin, unless the mother is also born in the Netherlands, in which case the father's country of birth is decisive.

Work and Benefits We have employment income data, for the job in which most hours are worked. For each employment contract, we have qualitative information on the working contracts of employed individuals.[10] We consider temp agency and 0-hour contracts as flexible, and code all other jobs as stable. We further merge data on welfare benefits, unemployment insurance benefits (UI), sickness benefits, and disability insurance (DI).[11] We collapse data on sickness and disability insurance together.

Education We match the results of the standardized test passed at 11-12 by Dutch children (*Cito Toets*) and the exam passed at the end of high school (*Eind Examen*). Data on the final exam is available from 2006 onwards, whereas data on the *Cito* test from 2007 onwards. We have information on individuals' educational attainment for all years in the sample, and registrations in an educational programme between 2003 and 2019.[12] The quality of these two datasets increases markedly between 2006 and 2014. We match each school's addresses to the corresponding neighbourhood centroid.[13] We compute the distance to school as the euclidean distance between the centroid of the residence place, and the school's neighbourhood centroid.

Crime The crime data come from Dutch the court records.[14] We observe all individuals going to court between 2001 and 2019, and can distinguish between five crime categories: property crimes, violent

[8] However, we identify a mistake in administrative records. On the 26th of February 2005, the municipality of Rotterdam added 25,000 individuals in one day, creating a spike in moving behaviour purely driven by a data error. We drop these observations for the analyses.

[9] See GBAPERSOON, KINDEROUDERTAB, GBAPARTNERBUS, and GBAHUISHOUDENS.

[10] See SECMWERKNDGAMNDBEDRAGBUS and BAANKENMERKENBUS.

[11] Stored in the SECMBIJSTMNDGAMNDBEDRAGBUS, SECMWERKMNDGAMNDBEDRAGBUS, and SECMZIEKTEMNDGAMNDBEDRAGBUS, respectively.

[12] See CITOTAB, EXAMVOTAB, ONDERWIJSINSCHRTAB, HOOGSTOPL.

[13] See BRINADRESSEN.

[14] see UITSROOMTAB and UITSTROOMRECHTERTAB.

and sexual crimes, traffic offences, drug offences, and other crimes. Using the individual identifiers, we can retrieve the place of residence of individuals who have committed a crime. We observe the municipality in which a given crime has taken place. We use criminals' place of residence in all analyses, in line with studies showing that criminals' place of residence is a key piece of information to understand spillovers in criminal behaviour (Damm and Dustmann, 2014).

4 Summary Statistics

Our units of analysis are individuals and administrative neighbourhoods (Dutch *buurten*), comparable in size to US Census tracts. Rotterdam had 569.985 inhabitants in 2005, and an average 6706 inhabitants per neighbourhood. It is a multicultural city: 33.2% of its inhabitants are born abroad, and 56.5% have a migration background (CBS, 2024). Map A5 shows that individuals with migration background sort into the central parts of the city.

4.1 Composition

Table 1 shows descriptive statistics on two samples. The first sample are all individuals aged 20 to 65 in treated neighbourhoods and the rest of the metropolitan area of Rotterdam and the Hague (henceforth RH). We exclude partially treated neighbourhoods in which not all streets were affected. In the full sample, treated neighbourhoods have a younger population, a higher share of individuals with non-Western origins, and a higher share of first and second generation migrants than the rest of the city. Moreover, treated neighbourhoods have a higher share of individuals who have committed a crime, and a higher share of individuals on welfare benefits. Treated areas have lower employment than the rest of the RH region, and display differences in the share of residents on benefits.

To have more balanced treated and control groups without relying on matching, we use unsupervised K-clustering to trim out neighbourhoods that are too dissimilar from treated areas. The second trimmed sample presents much better balance than the full sample. Table A3 presents an alternative trim, keeping only neighbourhoods in the first quartile of the housing prices distribution in the RH region.

Table 1: **Descriptive statistics on individuals in treated and control neighbourhoods in 2005**

	Full Sample				K-clustering Trim			
	Treated	Control	Norm. Diff	P-val	Treated	Control	Norm. Diff	P-val
Age	38.28	41.44	−0.253	0.000	38.28	39.04	−0.061	0.262
Male	0.53	0.50	0.054	0.000	0.53	0.51	0.033	0.006
Single parent	0.10	0.08	0.075	0.001	0.10	0.11	−0.037	0.130
Turkish	0.16	0.05	0.368	0.004	0.16	0.09	0.208	0.086
Moroccan	0.07	0.03	0.184	0.000	0.07	0.08	−0.011	0.809
Antillian	0.05	0.02	0.166	0.000	0.05	0.04	0.078	0.033
Surinamese	0.12	0.07	0.185	0.000	0.12	0.14	−0.033	0.397
Native	0.37	0.65	−0.577	0.000	0.37	0.43	−0.118	0.315
First Gen.	0.52	0.26	0.548	0.000	0.52	0.46	0.113	0.288
Second Gen.	0.11	0.09	0.064	0.005	0.11	0.11	0.003	0.886
Crime	0.02	0.01	0.087	0.000	0.02	0.02	0.025	0.039
Employed	0.60	0.68	−0.148	0.013	0.60	0.62	−0.033	0.589
Welfare	0.19	0.08	0.313	0.000	0.19	0.15	0.094	0.122
Unemployment	0.08	0.05	0.108	0.000	0.08	0.06	0.062	0.000
Sickness Ben.	0.10	0.09	0.057	0.004	0.10	0.10	−0.004	0.853
Obs. (N)	36,714	899,676			36,714	245,262		

Notes: This table presents the means of baseline characteristics for the treated and control groups for individuals aged 20-65 across the full sample and the K-clustering trimmed sample. The normalized difference (Imbens and Wooldridge, 2009) and the p-value of the difference in means are reported in the final two columns of each panel. Standard errors are clustered at the neighbourhood (*buurt*) level.

Figure 2 illustrates that treated neighbourhoods have higher residential turnover rates than the rest Rotterdam. Note that the black line represents the turnover in the city of Rotterdam. The share of in-and out-movers in treated neighbourhoods converge to the levels of the rest of the city by the end of the sample period. The left panel shows that the share of entrants in fully treated areas relative to the neighbourhood population is 18% at the time of treatment, decreasing to 15.4% in 2019. The share of out-movers in the right panel hovers at 15.3% for fully treated areas, declining to the level of untreated areas (11.6%) at the end of the sample. Treated areas see a decline in the portion of leavers around the time of treatment, whereas the portion of entrants remains stable, indicating that treated areas become slightly more populated. Pre-treatment trends in turnover are parallel.

Figure 2: **Share of the adult population moving in and out by treatment status. The black line only includes neighbourhoods in Rotterdam.**

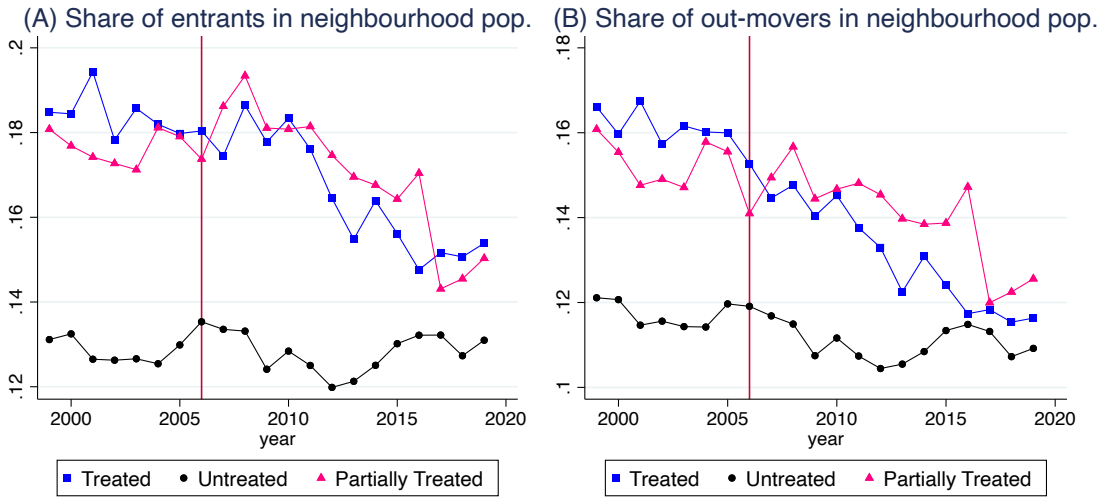


Figure 3 illustrates the evolution in neighbourhood composition over time. In pre-treatment years, the share of neighbourhood population in employment (panel A), the share on welfare (panel B), mean employment income (panel C), and mean welfare income (panel D) follow parallel trends. Around the time of the policy implementation (2005-2007), the share of individuals on welfare benefits decreases in treated and partially treated neighbourhoods (by 13.9% and 14.4%, respectively), whereas the corresponding share of individuals with employment income increases by 10.16% and 9.2%. Mean welfare spending decreases and employment income increases in treated neighbourhoods around in post-treatment years. Table A7 gives further information on the characteristics of individuals on welfare benefits in 2005 by their treatment status.

Figure 3: **Evolution in neighbourhood composition by treatment status**



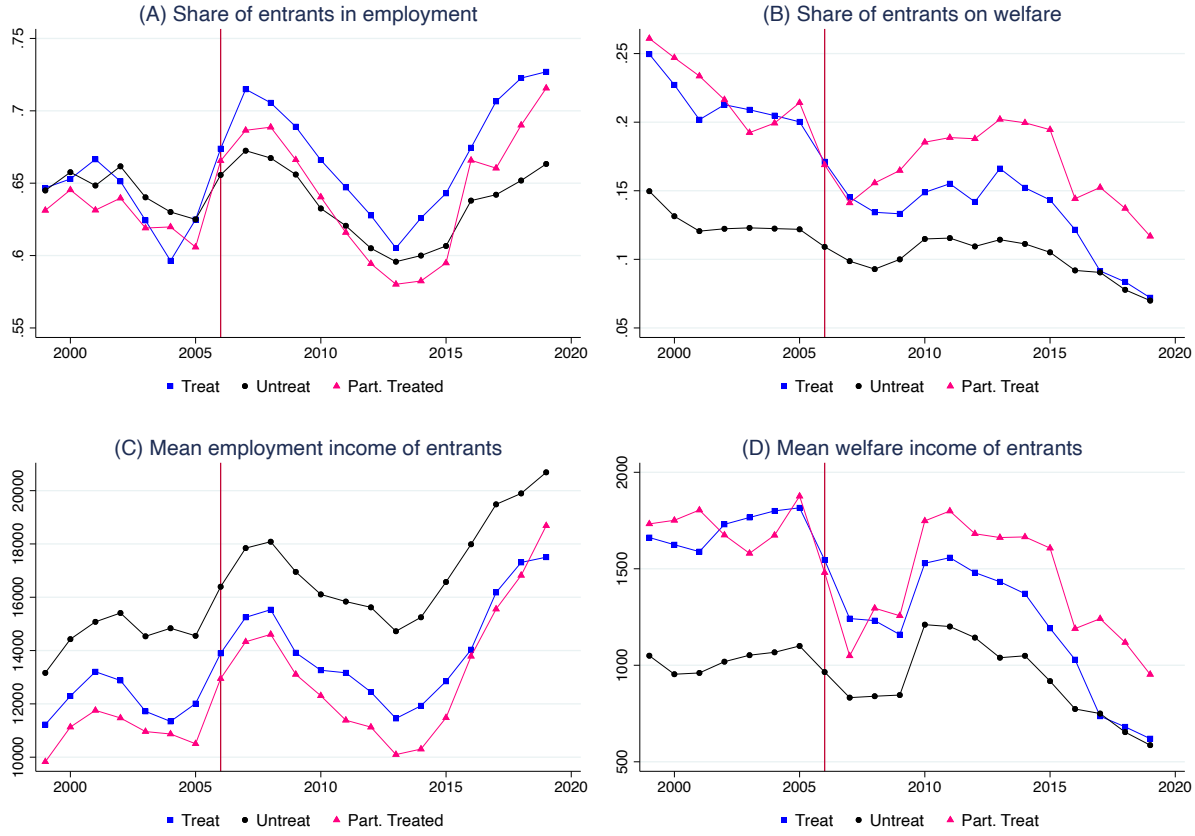
Note: Panel A: Share of neighbourhood population in employment. Panel B: Share of population on welfare. Panel C: Mean employed income. Panel D: Mean welfare income. The statistics are calculated by treatment status. The black line represents averages in the rest of the Rotterdam.

Figure A6 provides additional statistics on neighbourhood composition. Treated areas display no differ–ential changes in the share of residents receiving UI and DI benefits around the time of policy implementation. Treated areas show a steady increase in the share of flexible workers (temp-agency or zero-hour contracts), and single-parent households. The ethnic composition of treated neighbourhoods remains stable, whilst the share of residents who have committed a crime declines by half over the sample period.

4.2 Inflows and outflows

Figure 4 illustrates changes in the inflow of residents to untreated, partially, and fully treated neighbourhoods. The inflow is the main margin through which the policy operates, preventing individuals without employment income from entering treated neighbourhoods. Panel A shows a decrease in the share of entrants receiving welfare benefits, and an increase in the share of entrants with employment income. These changes are substantial in relative terms: between 2005 and 2007 they imply a 27.5% decrease in the share of entrants on welfare benefits, and an 14.4% increase in the share of employed entrants. The mean employment income of entrants follows the trend in other neighbourhoods, whereas welfare benefits claims of entrants decrease markedly relative to the rest of the city. The trends are parallel in pre-treatment periods.

Figure 4: **Evolution in the composition of entrants in neighbourhoods by treatment status**



Note: Panel A: Share of entrants receiving welfare benefits. Panel B: Share of employed entrants. Panel C: Mean employment income of entrants. Panel D: Mean employment income of entrants.

In Figure A8, we show that the share of entrants on UI or DI benefits does not change significantly around the treatment time, suggesting the law affected entrants receiving welfare benefits more than entrants on UI or DI benefits. Figure A8 further indicates that the share of entrants who have committed a crime decreases, whereas the share of entrants with flexible work contracts, and that of single parents grow smoothly around treatment time. The latter two facts rule out the possibility that entrants circumvented the policy by entering treated neighbourhoods with a partner or with a zero-hour contract without engaging in actual work. The ethnic composition (measured by the country of origin) of entrants is constant throughout the sample, except for an increase in native Polish entrants and a secular decrease in native entrants.

Figures A8 and A10 demonstrate that inflow and outflow composition almost perfectly track each other. Out-movers are no more likely to receive UI or DI benefits around the treatment time, but are more likely to hold flexible work contracts. The data show no evidence of a 'majority flight' as in Card et al. (2008). Rather, the share of native entrants and leavers decreased slowly throughout the sample. Interestingly, Figure A9 shows that the share of leavers on welfare decreased around the introduction of the policy, whereas the share of leavers with employment income increased. The former effect is consistent with a behavioural response to the policy by incumbent residents, choosing to stay put and find a job instead of moving and being exposed to the policy restrictions.

Based on data from Lam et al. (2025), we map the quality of residential moving flows both before and after treatment. Figure A11 plots, for each neighbourhood, the average household income of children whose parents ranked between the 20th and 30th percentiles nationally (see Chetty and Hendren, 2018). We compute these means separately for the neighbourhoods that treated individuals move out of and for those they move into. The plot aims to determine whether individuals moving in (out) move into higher or lower opportunity neighbourhoods. The data suggest that the policy succeeded in attracting residents from slightly better-off neighbourhoods in 2005, although the differences are small. Overall, people continue moving into and out of similar quality neighbourhoods after the policy.

4.3 Incumbents sample

To evaluate the indirect effects the policy, we restrict our sample to long-term residents (the incumbents). We define incumbents as individuals having lived at least 2 years in the same neighbourhood before treatment. For the triple difference specification, we keep only individuals aged 20 to 45 in the sample to obtain stronger covariate balance - especially in terms of age - across the 6 year tenure threshold.

Table 2: **Descriptive statistics on incumbents in treated and control neighbourhoods with ± 6 years of residency**

	Treated		**Control**			
	- 6 years	**+ 6 years**	**- 6 years**	**+ 6 years**	**P-val DiD**	**Norm. DiD**
Age	30.09	33.69	30.28	34.17	0.288	−0.045
Male	0.53	0.51	0.52	0.49	0.474	0.027
Single head hh	0.09	0.13	0.08	0.16	0.000	−0.168
Turkish	0.11	0.23	0.06	0.10	0.007	0.275
Moroccan	0.08	0.09	0.06	0.09	0.100	−0.069
Antillian	0.04	0.03	0.06	0.03	0.002	−0.093
Surinamese	0.11	0.14	0.08	0.13	0.068	−0.068
Native	0.28	0.34	0.40	0.44	0.507	0.051
First Generation	0.64	0.51	0.49	0.39	0.214	−0.079
Second Generation	0.08	0.15	0.11	0.16	0.180	0.050
Criminal Record	0.03	0.02	0.02	0.02	0.100	−0.054
Employed	0.64	0.68	0.69	0.71	0.384	0.039
Welfare	0.17	0.18	0.12	0.16	0.101	−0.091
Unemployment Insurance	0.08	0.09	0.05	0.08	0.023	−0.050
Sickness Benefits	0.06	0.08	0.04	0.07	0.331	−0.020
Observations (N)	5,647	14,629	15,484	43,704		

Notes: This table presents the means of baseline characteristics for incumbents in the treated and control groups comparing individuald with −6 years and +6 years of housing tenure in 2005. The sample includes incumbents aged 20-45 in the K-clustered sample. The p-value of the difference-in-differences (DiD) and the normalized DiD (Imbens and Wooldridge, 2009) are reported in the final two columns. Standard errors are clustered at the neighbourhood (*buurt*) level.

Table 2 shows descriptive statistics for the treated and control groups in 2005, separating between in–

dividuals above and under the 6 year tenure threshold in the K-clustered sample.[15] The p-value and the normalized differences indicate that the covariate differences between treated and control are similar in treated and control groups. Table A5 shows the same table for the trimmed sample, excluding neighbourhoods above the first quartile of housing prices.

5 Empirical strategy

5.1 Sample trim through K-clustering

As evident in the summary statistics, the full unrestricted sample exhibits poor covariate balance. Treated neighbourhoods are less affluent and more diverse than the average neighbourhood in the RH region. To trim the sample without using matching, we apply K-means clustering — an unsupervised learning algorithm — to all baseline covariates. We then restrict our control sample exclusively to untreated units that fall into clusters containing treated units, effectively trimming dissimilar controls. We specify 30 clusters for our analysis, though our estimates are robust to alternative cluster counts.

We propose an alternative trimming method by simply restricting the sample to all neighbourhoods in the RH region that fall into the first quartile of the neighbourhood price distribution. The results are virtually unaltered when using the housing price trim instead of the K-clustering.

5.2 Difference-in-differences

We estimate the following event-study model at the individual level

$$Y_{i,n,t} = \alpha_t + \eta_n + \phi_i + \sum_{k=-4}^{-2} \beta_k^{pre} \left[\mathbb{1}\{G_i = g\} \cdot \mathbb{1}\{t = k\}\right] + \sum_{k=0}^{9} \beta_k^{post} \left[\mathbb{1}\{G_i = g\} \cdot \mathbb{1}\{t = k\}\right] + \varepsilon_{i,n,t} \tag{1}$$

where $Y_{i,n,t}$ is the outcome of interest for individual i in neighbourhood n at time t, and β_t is the time variant policy effect. The first summation term estimates pre-treatment coefficients, and the second summation term estimates post-treatment coefficients. $\mathbb{1}\{G_i = g\}$ is an indicator taking the value $g = 1$ for individuals in treated neighbourhoods, and 0 otherwise. $\mathbb{1}\{t = k\}$ equals 1 if time period t is equal to period k, and 0 otherwise. We set $k = 0$ for the year 2006, and normalize $\beta_{-1} = 0$, such that 2005 serves as the reference year. α_t are year fixed effects, η_n neighbourhood fixed effects, ϕ_i individual fixed effects, and $\epsilon_{i,n,t}$ is the error term.

We estimate two variations of Equation (1). Firstly, we run the model on the full (non trimmed) sample of residents, without distinguishing between stayers and movers. This specification yields the total effect of the policy, summing the indirect effect on long-term residents with the direct effects due to compositional changes. To avoid absorbing relevant variation, we use neighbourhood and year fixed effects, omitting individual fixed effects (ϕ_i) in the estimation. In the second variation, we keep all residents staying in the same neighbourhood in the two years before treatment (the incumbents). We then determine incumbents' treatment status using their place of residence those years ($t = \{-1, -2\}$). This specification can be interpreted as an intention-to-treat (ITT) for individuals living in the same neighbourhood in the two years before treatment. The ITT specification captures both the indirect effects of the policy on incumbents and the compositional effects resulting from their moves in and out of treated neighbourhoods after 2005 ($t = -1$).

[15] See Tables A8 for the 6-year table for the full sample (no incumbency restrictions and no trim), and Table A9 for incumbents (without trim). See A6 for the table excluding control neighbourhoods in R'dam.

The second model is estimated with individual and year fixed effects on the K-clustered sample. Table A4 shows some summary statistics for the treated and control group of incumbents.

We estimate Equation (1) on employment, work income, welfare income (*bijstand*), UI income, DI/sickness income, having some source of income, and criminal behaviour. All regressions are run in levels, and we explicitly separate between intensive and extensive margins for labour market outcomes. The latter avoids functional form issues related to 0-outcomes (Chen and Roth, 2024). Standard errors are clustered at the neighbourhood level.

5.3 Triple-difference

The specification in Equation 1 relies on the assumption of parallel trends and no confounding treatments. Although we argue that these are likely hold, the unique setting of the RW allows us to run a third difference which presents two main advantages. Firstly, it strengthens our results by suppressing concerns on unob–served confounders affecting our estimates. Secondly, we can disentangle the 'peer' effect of the policy from the incentive effect of the policy, which should only affect those with less than 6 years of housing tenure. We therefore estimate the following triple-difference (DDD) model:

$$\begin{aligned} Y_{i,n,t} = \alpha_t + \eta_n + \phi_i \\ + \sum_{\substack{k=-4 \\ k\neq -1}}^{9} \gamma_k \left[\mathbb{1}\{G_i = g\} \cdot \mathbb{1}\{t = k\}\right] + \sum_{\substack{k=-4 \\ k\neq -1}}^{9} \delta_k \left[S_i \cdot \mathbb{1}\{t = k\}\right] \\ + \sum_{k=-4}^{-2} \beta_k^{pre} \left[\mathbb{1}\{G_i = g\} \cdot S_i \cdot \mathbb{1}\{t = k\}\right] + \sum_{k=0}^{9} \beta_k^{post} \left[\mathbb{1}\{G_i = g\} \cdot S_i \cdot \mathbb{1}\{t = k\}\right] \\ + \varepsilon_{i,n,t} \end{aligned} \tag{2}$$

with γ_k being the time specific interaction between the treatment and the time dummies, and δ_k being the interaction between the six threshold S_i and the time dummies. Note that the interaction between the treatment $\mathbb{1}\{G_i = g\}$ and the six year rule S_i is absorbed by the individual FE, hence the omission. The DDD specification compares individuals above and below the 6 year residence tenure threshold (fixed in 2005) in treated and control neighbourhoods. We run the specification on the sample of incumbents (aged 20-45) in the K-clustered sample and the alternative trimmed sample.

To examine potential spillovers, we exclude residents in treated neighbourhoods and recode the treatment variable such that directly adjacent neighbourhoods are coded as treated. We then reconstruct a sample using K-means clustering on the spillover neighbourhoods, and re-run the same specification. As a placebo exercise, we implement this approach on a randomly selected set of neighbourhoods in the RH region. Figure A7 shows that none of the neighbourhoods exhibits a sudden change of the share of residents on welfare, except from Katendrecht.[16] We exclude this latter neighbourhood and all partially treated neighbourhoods from all analyses.

5.4 Within-neighbourhood variation in exposure

Access to the individual-level school data prior to 2006 is restricted due to privacy restrictions. We therefore rely on within-neighbourhood variation in policy exposure to estimate its effects on children. Following Carrillo et al. (2023), we identify the effect of additional years of policy exposure by exploiting age differences

[16] This small neighbourhood was almost entirely revamped around 2010, with strong effects on neighbourhood composition.

at the time of implementation. For instance, children who were younger at the time of policy implementation were exposed to the policy for a greater number of years by the time they took their exams. Formally, we calculate exposure as

$$E_{it} = (\bar{B} - \mathrm{A}_{i,2006}) \times \mathbb{1}[\mathrm{A}_{i,t} \leq \bar{B}] \tag{3}$$

A_{it} is the age of student i at time t, $\bar{B}$ is the age at which a student takes the relevant exam (11-12 for the Cito test, and 15-18 for the final exam). The Cito test is taken around the middle of the school year (January-February), we therefore only observe exam scores for younger children from 2007 onwards.

Note that we drop the subscript n because we leverage only within (treated) neighbourhood variation. For the Cito test score, we measure exposure based on place of residence of children. In the Netherlands, primary schools are geographically more accessible than secondary schools (CBS, 2022b). Children typically attend a primary school in their neighbourhood, but may travel farther for their secondary schooling. For the final exam, we therefore use the place where children are schooled rather than the place of residence to compute exposure length. We further distinguish between higher and lower educational tracks in high school, which differ in both the age of examination and the type of exam administered. Test scores are standardized, so estimated effects can be interpreted as fractions of a standard deviation of the pooled grade distribution across years.

Leveraging within-neighbourhood across-cohort variation, we estimate the effects of each additional year of exposure to the policy on the outcomes of interest. The regression equation can be written as

$$Y_{i,t} = \beta_t E_{i,t} + \mathbf{X}'_{i,t}\Omega + \epsilon_{i,t} \tag{4}$$

The subscript i stands for children, t for time. Y_{it} is the outcome of interest, $\mathbf{X}'_{i,t}$ is a vector of controls including gender, migration generation, household type, and ϵ_{it} is the error term. The coefficients of interest β_t can be interpreted as the post-2006 cohort fixed-effects within treated neighbourhoods. We focus on three main outcomes: test scores for the Cito test, the exit exam, and the probability of participating in the exit exam.

5.5 Heterogeneous effects using causal forests

The richness of the data allows us to investigate heterogeneity in the estimates using causal forests (Athey et al., 2019; Athey and Imbens, 2019; Athey and Imbens, 2017). Causal forests are a relatively recent development in the supervised machine learning literature. They enable a data-driven heterogeneity analysis, capable of uncovering non-linearities and interactions that would otherwise be overlooked.

The goal of causal forests is to estimate the Conditional Average Treatment Effect (CATE) given by

$$CATE = \mathrm{E}\left[Y_{1i} - Y_{0i} | X_i = x\right] \tag{5}$$

where Y_{1i} and Y_{0i} are the potential outcomes of treated and untreated individuals, and X is a vector of individual characteristics.

The intuition behind the regression trees underlying causal forests is to make splits in the data based on the values of the covariates. At each node, the algorithm performs the split that (locally) leads to the greatest improvement in the objective function (Baiardi and Naghi, 2024). These split-out groups are referred to as *leaves*. The *tree* is defined when no more splits in the dataset are possible given the

algorithm's stopping function. To investigate heterogeneous causal effects, Athey and Imbens (2016) propose to maximize a criterion function that maximizes across-leaves variance, and minimizes within-leaf treatment effect variance.[17] To avoid overfitting, the sample is split in two equal-sized subsamples. The first sample is used to determine the splits, whilst the second is used to predict the CATE for the covariates we include. Athey et al. (2019) call this the 'honest approach'.

We closely follow Athey and Wager (2019) for the implementation, and cluster estimation at the neighbourhood level. The four outcomes of interest are employment, welfare benefit take-up, other benefits take-up, and having a source of (legal) income, all at the extensive margin. For each individual and outcome, we estimate the CATE based on age, gender, migration generation, migration background, household type, employment income (intensive), benefit(s) income, and criminal behaviour. We run the causal forest on the triple difference (K-clustered) sample of incumbents. For ease of interpretation and implementation, we take out the dynamic structure in Equation (2), yielding a canonical 2x2 DiD model.

6 Results

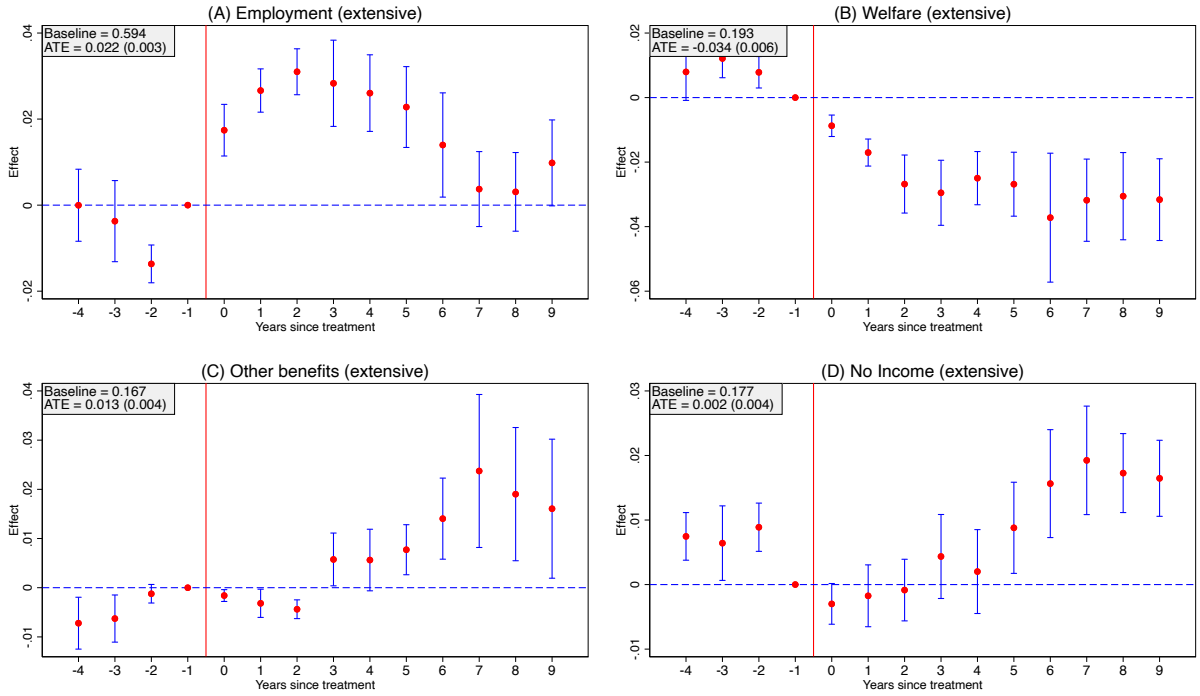
6.1 Compositional effects

The first set of results shows the total effects of the RW on individual outcomes in treated neighbourhoods, as in the first variation of Equation (1). In this specification, individuals in treated neighbourhoods are compared to individuals in untreated neighbourhoods (thus excluding partially treated areas). The estimated effect is the sum of the direct compositional effect and the indirect effect on incumbent residents relative to the baseline year.

Figure 5 presents the total effect of the policy on employment (panel A), welfare benefits (panel B), other benefits (panel C), and on having no registered income (panel D) at the extensive margin. The baseline is the 2005 outcome mean in *untreated* neighbourhoods. The policy increased employment in treated neighbourhoods by 3.2 percentage points (5.4%), and reduced welfare benefits take-up by 2.2 pp (11.4%) 2 years after treatment. The share taking up other government benefits (illness/disability and UI benefits) does not increase significantly in the short-term, but increases in the medium term. This increase is a mechanical bi-product of the policy: more people in targeted neighbourhoods start working, and automatically become eligible to worker insurance programmes. We do not detect much movement on (legal) income take-up. The effects on welfare persist 9 years after treatment, although the employment effects seem to fade out after 6 years.

[17] The *forest* aggregates CATE estimates across trees. The procedure to make forests is analogous to the procedure for trees. Once tree k is grown, tree $k+1$ is grown on a slightly different subsample, until the ensemble of trees reaches another stopping criterion. At the stopping point, the forest is defined. Note that these are decorrelated trees.

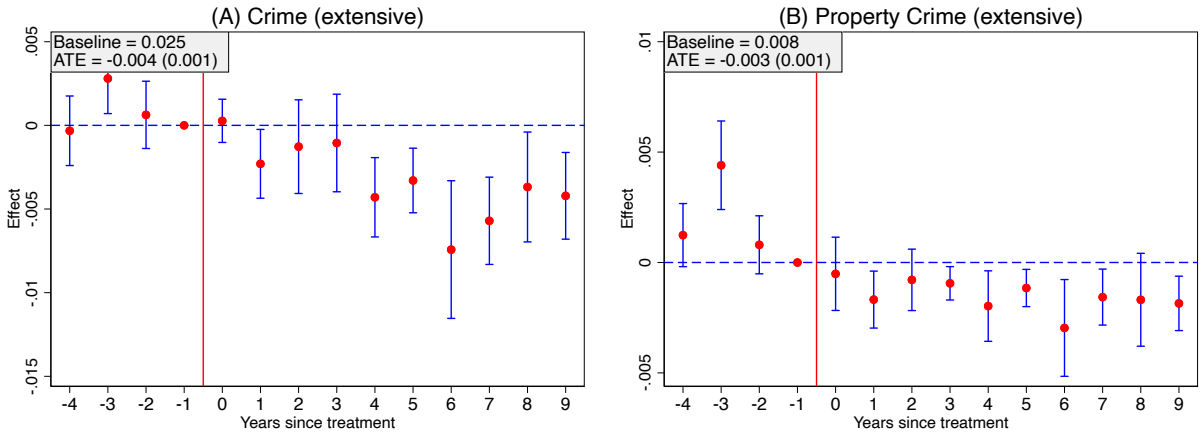
Figure 5: **Event study DiD on labour market outcomes - compositional effect**



Note: The plots show the results of estimating Equation (1) on employment (panel A), welfare benefits (panel B), other benefits (panel C) , and no income (employment & transfers - panel D), at the extensive margin. All adult individuals living in fully treated neighbourhoods are treated. All adults living in untreated neighbourhoods (non partially treated) and outside of Katendrecht are the controls. The baseline is the outcome mean in treated neighbourhoods in 2005 ($t = -1$).

Figure 6 shows the total effect of the policy on residents' the probability of committing a crime (panel A), and the share of single-parents households (panel B). The share of residents having committed a crime decreases by 0.5 pp (25%) five years after treatment, with negative though insignificant estimates in the three years following the policy. By the end of the sample, the share of residents who committed a crime in a given year decreases by a third relative to baseline, indicating strong persistent effects. The effects on property crimes are of similar magnitude, although parallel trends do not hold as nicely. The share of 1st and 2nd generation migrants in the neighbourhoods is unaffected by the policy (not provided in the figure).

Figure 6: **Event study DiD on crime - compositional effect**



Note: The plot show the results of estimating Equation (1) on the probability of committing a crime. All adult individuals living in fully treated neighbourhoods are treated. All adults living in untreated neighbourhoods (excluding partially treated) and outside of Katendrecht are the controls.

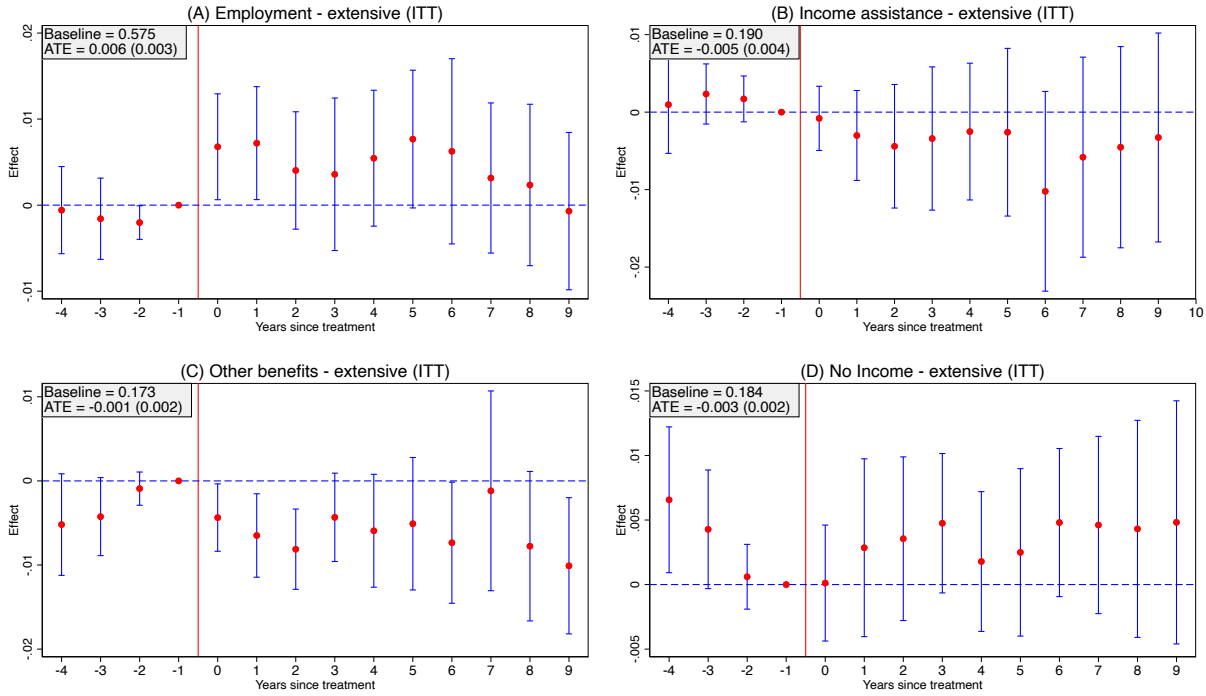
Overall, the policy increased the share of residents in employment, whilst reducing the share of residents on employment benefits. The share of residents having committed a criminal offence decreases four years after treatment. The effects on employment, welfare, and crime are sizeable, especially when compared to the 'ghetto policy' enforced in Denmark, which, although similar to the RW, induced smaller changes in neighbourhood composition and did not include any displacement threats (Govind et al., 2025).

6.2 Indirect effects - long-term residents

To measure the indirect effects of the policy on long-term residents, net of the compositional effects, we restrict the sample to all individuals living in the same neighbourhood in the three years before treatment. These individuals are hereafter referred to as the *incumbents*. Table A4 summarizes the main features of the sample. The estimates can be interpreted as an intention-to-treat (ITT), acknowledging that some incumbents may relocate following treatment.

Figure A14 shows the effects of the policy on incumbents' labour market outcomes at the extensive margin. Panel A shows a positive effect (0.8 percentage points, 1.4%) of the policy on employment 2 years after treatment. The effect persists 5 years after the introduction of the policy, before becoming insignificant. Panel B shows a negative though insignificant effect on the probability of taking up welfare 2 years after treatment (0.5 pp, 2.6%). Panel C shows a contemporaneous decrease in the take-up of DI or UI benefits (0.7 pp, 4%). Panel D could indicate a decrease in the share of incumbents not declaring an income, although the estimates are noisy.

Figure 7: **Event study DiD on incumbents' labour market outcomes - extensive margin**



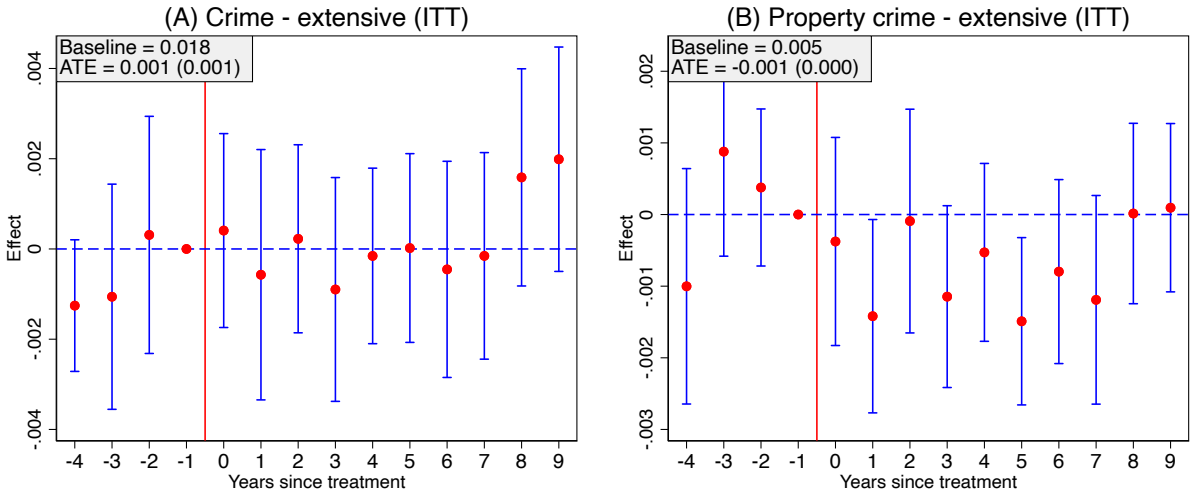
Note: The plots show the results of estimating Equation (1). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on declaring an income. The baselines are the 2005 variable mean for incumbents in *untreated* neighbourhoods.

Figure A13 shows the effects of the policy on labour market outcomes at the intensive margin. In post-treatment years income decreases (panel A), whereas welfare income, other benefits, and total income stay stable (panels B,C,D).

The policy's impact on employment is roughly half the size of that of an active labour market programme. In their meta-analysis, Card et al. (2018) estimate that the average short-run impact of these programmes between 1 and 3 pp, whereas the medium-run effects hover around 2 to 5 pp. Our result of a 0.8 pp increase in employment 2 years after treatment is therefore non-trivial in magnitude, although it is much smaller than some more successful activation programmes (e.g. Markussen and Røed, 2016).

The effects of the RW on welfare participation are smaller in magnitude than the expected effect from an activation programme for individuals out of employment. Card and Hyslop (2005) estimate a 14 pp short-term decrease in welfare dependence due to the incentives provided by the SSP programme, though the effect vanishes by the end of the programme (after 69 months). Van der Klaauw and Van Ours (2013) estimate the effects of a bonus-penalty activation programme targeted at welfare recipients in Rotterdam. They find that the bonus incentive increases the exit rate from welfare by about 2 pp, whereas the penalty incentive decreases the exit rate by 5 pp. The 0.8 pp decrease in welfare take-up we observe is modest compared to targeted programmes, but non-trivial considering it results from a policy not directly targeting welfare participation.

Figure 8: **Event study DiD on crime for incumbents**



Note: The plots show the results of estimating Equation (1). Panel A: effects on probability to commit a crime. The baseline is the 2005 variable mean for incumbents in *treated* neighbourhoods.

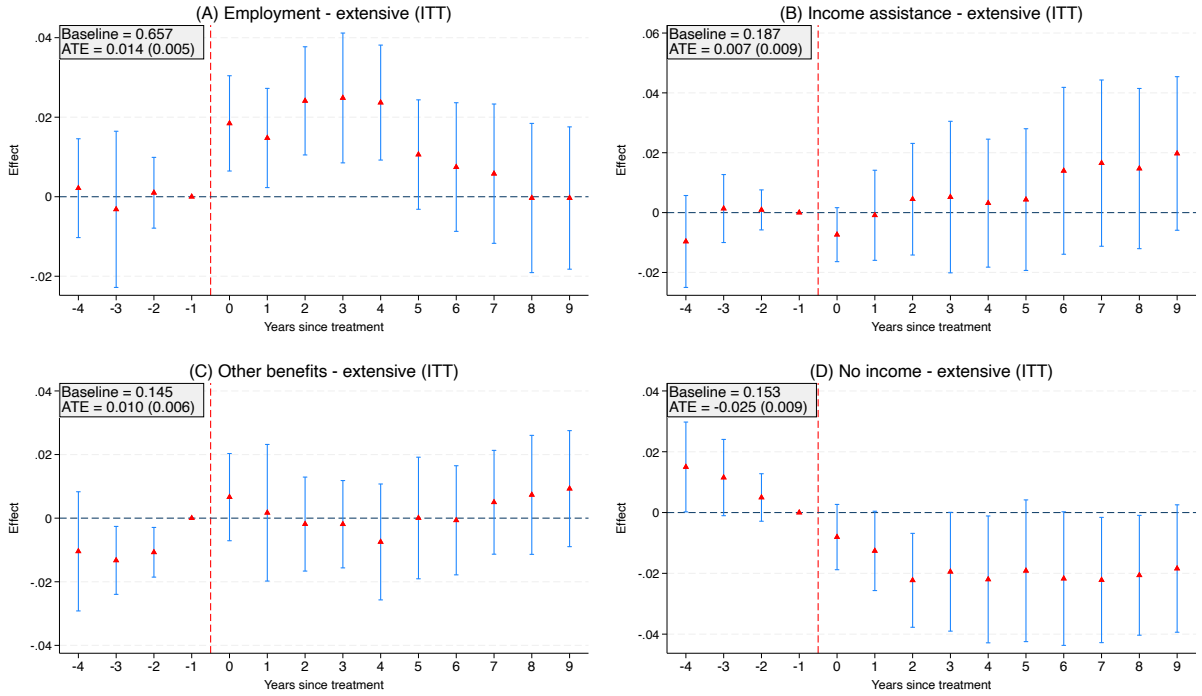
Figure 8 shows the effects of the policy on crime among incumbents. Five years after the implementation of the policy we find no reduction in the share of incumbents committing a crime, although there seems to be a reduction in property crimes.

Overall, the estimated effects for incumbents are consistent with two broad mechanisms. First, incumbents may have strong idiosyncratic preferences for living in treated neighbourhoods. In order to continue residing in those neighbourhoods after a forced or voluntary move, incumbents may react to the policy by taking a job. Second, a localized change in composition may induce incumbents to work and rely less on welfare payouts through peer effects. The change in composition would then act as a break in local welfare culture. A similar mechanism could affect crime rates: incumbents have less contacts with criminals, leading to a decrease in criminal offences.

6.3 Indirect effects - triple difference

To strengthen the credibility of our identification relative to the standard difference-in-differences, we estimate a triple-difference specification (Equation 2). This approach improves upon the standard DiD by netting out unobserved, group-specific trends that might otherwise violate the parallel trends assumption. Importantly, this strategy also allows us to isolate the policy's direct incentive effects from potential peer effects driven by changes in neighborhood composition. Our estimates capture these incentive effects, but do not rule out peer effects entirely, which we plan to explore in future work using street-level data.

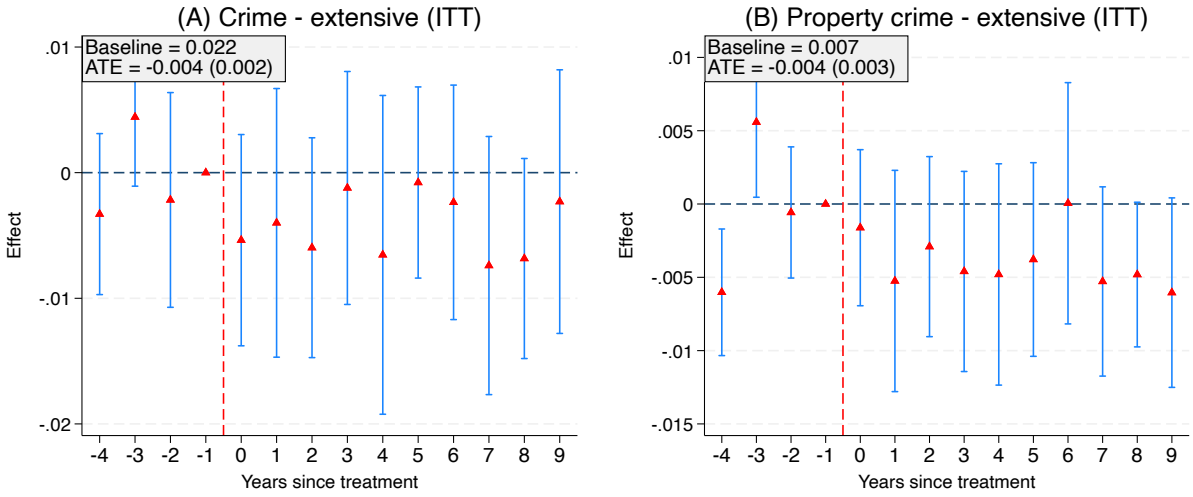
Figure 9: **Event study DDD on incumbents' labour market outcomes - extensive margin**



Note: The plots show the results of estimating Equation (1). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on having no income. The baselines are the 2005 variable mean for incumbents in *untreated* neighbourhoods.

The triple difference estimates in Panel A of Figure 9 show a 1.8 pp (2.7%) effect on employment and a 2.1 pp (13.7%) decrease in having no registered income (Panel D). Welfare and other benefits adoption do not change much (Panels B & C).

Figure 10: **Event study DDD on crime for incumbents**



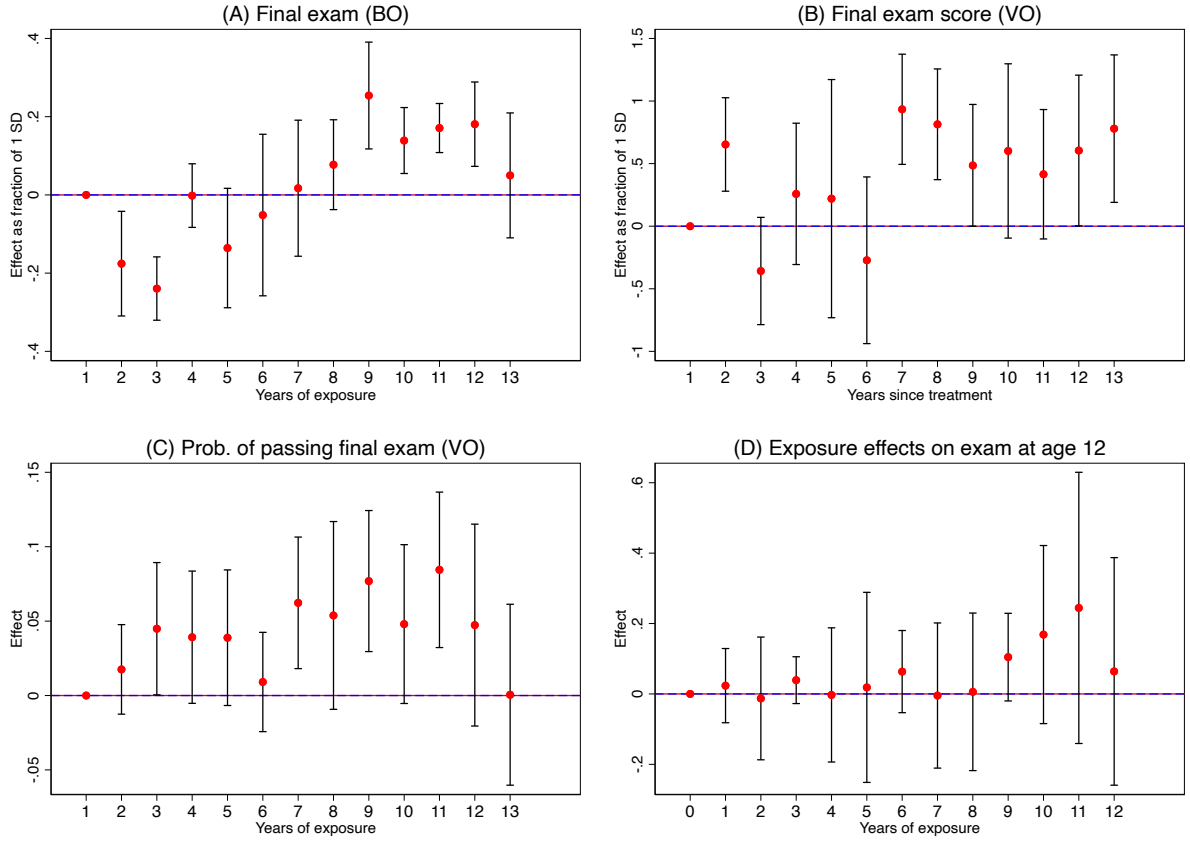
Note: The plots show the results of estimating Equation (2). Panel A: effects on probability to commit a crime. Panel B: effects on probability to commit a property crime. The baseline is the 2005 variable mean for incumbents in *untreated* neighbourhoods.

Figure 10 shows the effects of the RW on long term residents with less than 6 years of residency tenure. We find negative, although imprecisely estimated, effects both on all crimes (0.5pp, 18.2%, Panel A) and property crimes (0.5pp, 71%, Panel B). Our results on crime are sizeable: the magnitudes are almost half of that of successful targeted programmes to reduce violence (Heller et al., 2017).

6.4 Direct effects - youth

To estimate the effects on schooling outcomes, we rely on within-neighbourhood variation in exposure. The baseline is the cohort of children who took their exams in 2006, having had 0 or 1 year of policy exposure depending on the exam they take. We estimate the effects on children taking their Cito test (age 11-12) and teenagers taking their final exam (age 15-18). We distinguish between higher and lower educational tracks, and test scores are standardized. Estimated effects can be interpreted as fractions of a standard deviation of the pooled grade distribution across years.

Figure 11: **Effects of exposure on youth schooling outcomes**



Note: Results of estimating Equation 4 leveraging within-neighbourhood variation in the length of exposure to the RW at the moment of taking the exam. Panel A shows the exposure effects on test scores for students taking the final exam in lower tracks (age 15-18). Panel B shows the exposure effects for students taking the final exam in the higher tracks (age 17-18). Panel C shows the exposure effects on the probability of passing (grade > 5.5) on the final exam for teenagers in the higher tracks. Panel D shows exposure effects on the scores on the Cito test taken at 11-12.

Panel A of Figure 11 shows the exposure effects on test scores in lower tracks (age 15-18), and panel B for higher tracks (age 17-18). Panel C shows the effects of the policy on the probability to pass the exam for students in the higher tracks (VO). Panel D shows exposure effects on the scores on the *Cito* exam taken at 11-12.

The four estimates are noisy, and the profile of the figures is sensitive to the chosen baseline. For the lower track students (panel A), the profile of the exposure effects is broadly consistent with human capital theory. The pattern does not hold for higher track students (panel B), although these students see an increase in the probability to pass the final exam (panel C). Panel D indicates null effects among younger students.

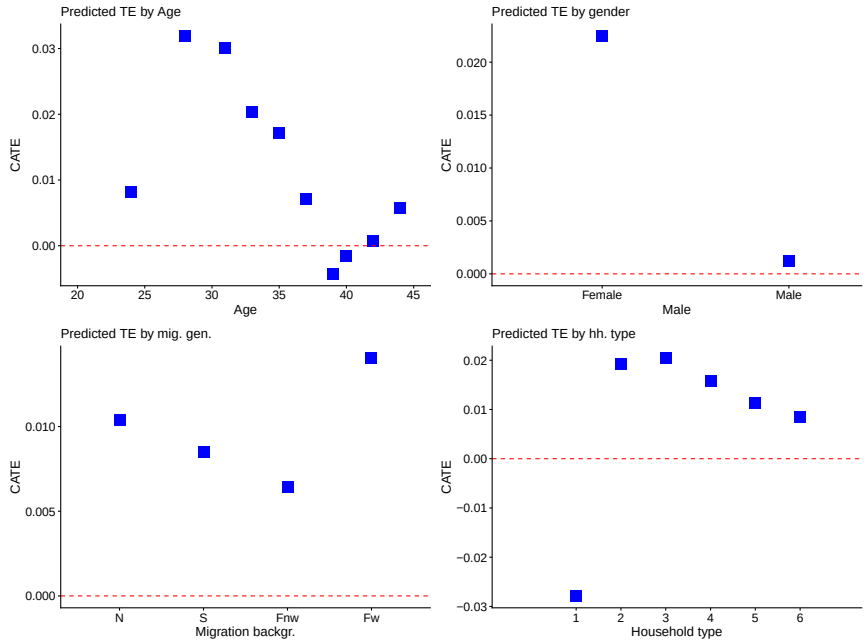
The results for youth outcomes are inconsistent across schooling levels and ages. We therefore caution against making inferences based on them. Note that these findings are consistent with early studies on the moving to opportunity (MTO) experiment (e.g. Fryer and Katz, 2013), showing that place effects do not have medium-run impacts on human capital accumulation and labour market outcomes for young individuals.

6.5 Heterogeneity - incumbent adults

We investigate heterogeneity in treatment effects for incumbent residents using causal forests. As explained in Section 5, we run a simple 2x2 DDD, thus taking out the dynamic dimension in Equation (2). We include age, gender, migration generation, migration background, household type, employment income (intensive), benefit(s) income, and criminal behaviour in the estimation. The sample is composed of all individuals in the K-clustering trimmed sample aged between 20 and 45 living in the same neighbourhood in the two years before the policy is enforced). As in Athey and Wager (2019), we present the CATE for the variables that have and above average 'importance' for the data splits operated by the algorithm.

Running the causal forest on the incumbents sample with employment as outcome yields $ATE = X \quad (SE = X)$, which is higher than the ATT estimate in the event study DiD specification. We chose the specification by including covariates that could plausibly predict the outcome and treatment status. We grow 500 trees per model, with results showing little sensitivity to this parameter.[18]

Figure 12: **CATE on employment by characteristic - extensive margin**



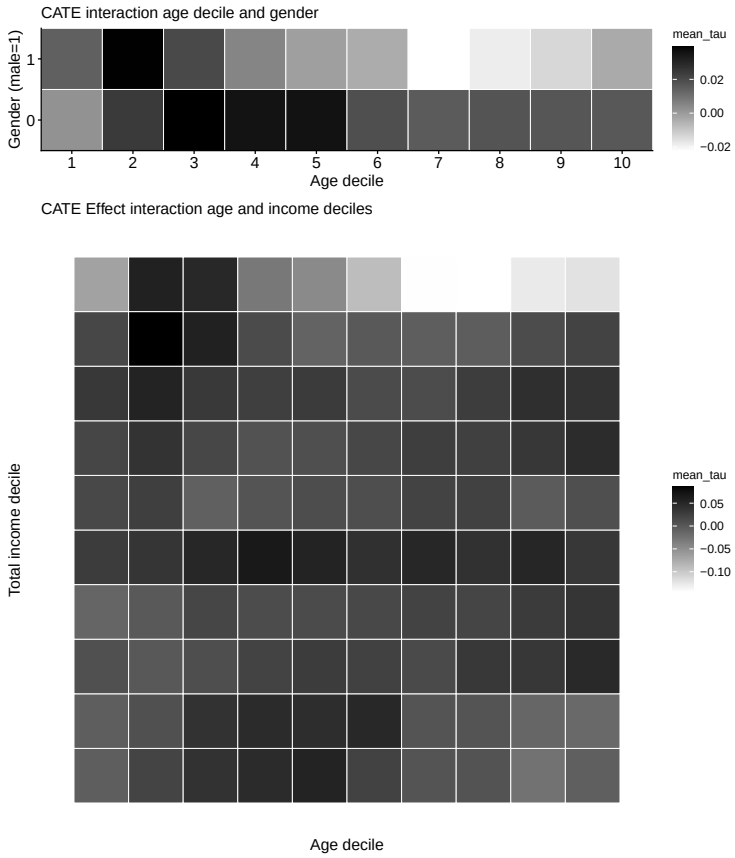
Note: Result of estimating 2x2 DiD setting on incumbents sample using causal forests. Panel A: mean CATE by age decile. Panel B: mean CATE by gender. Panel C: mean CATE by migration background [Natives, Second Gen, First Gen non-western, First Gen western]. Panel D: mean CATE by household type. Household types: (1) single person household; (2) non-married couple & no kids (3) married couple & no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

Figure 12 shows the mean CATE by age decile in the data (panel A), gender (panel B), migration generation (panel C), and by household type (panel D). Household types are classified as (1) single person household; (2) non-married couple & no kids (3) married couple & no children; (4) non-married with kids; (5) married with kids; (6) single parent with children. The CATE for employment shows that predicted conditional treatment effects are significantly higher for women, and individuals between the age of 20 and 35 years. Households with children react relatively more than households without children, consistent with

[18]Growing more trees is computationally heavy. We have tried running multiple forests and then aggregating them together to increase the number of tress grown. The results are virtually unchanged by this procedure.

the idea that less mobile individuals react more to the displacement threat. The estimated CATE is similar for natives, 1st, and 2nd generation migrants.

Figure 13: **CATE on employment by pairs of characteristics - extensive margin**

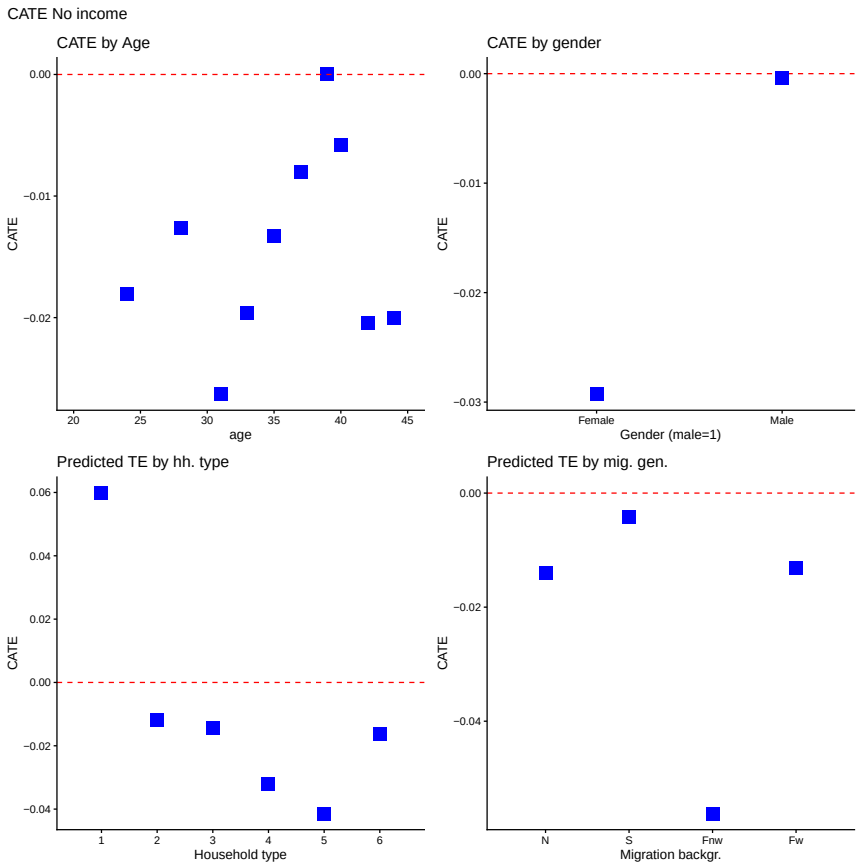


Note: CATE on employment by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

Figure 13 explores interactions between dimensions of heterogeneity. The upper panel shows the interaction between age decile and gender. The lower panel shows the interaction between age decile and total income deciles. Darker shades indicate larger CATE magnitudes. The heatmap shows that effects are largely driven by women between the 2nd and 7th decile of the age distribution. Younger and less affluent individuals seem to react more to the policy. The results of the heterogeneity analysis are consistent with evidence from Card et al. (2018), suggesting that labour market activation programmes have stronger effects on women, and modest effects for older individuals.

Re-running the causal forest with the same covariates, on the sample sample, with a dummy for having no registered income as outcome variable yields $ATE = X \quad (SE = X)$. The results of the estimation are shown in Figure 14. Much like the results for employment, the result ATE on welfare participation is larger in magnitude compared to the event-study specification.

Figure 14: **CATE on no income - extensive margin**

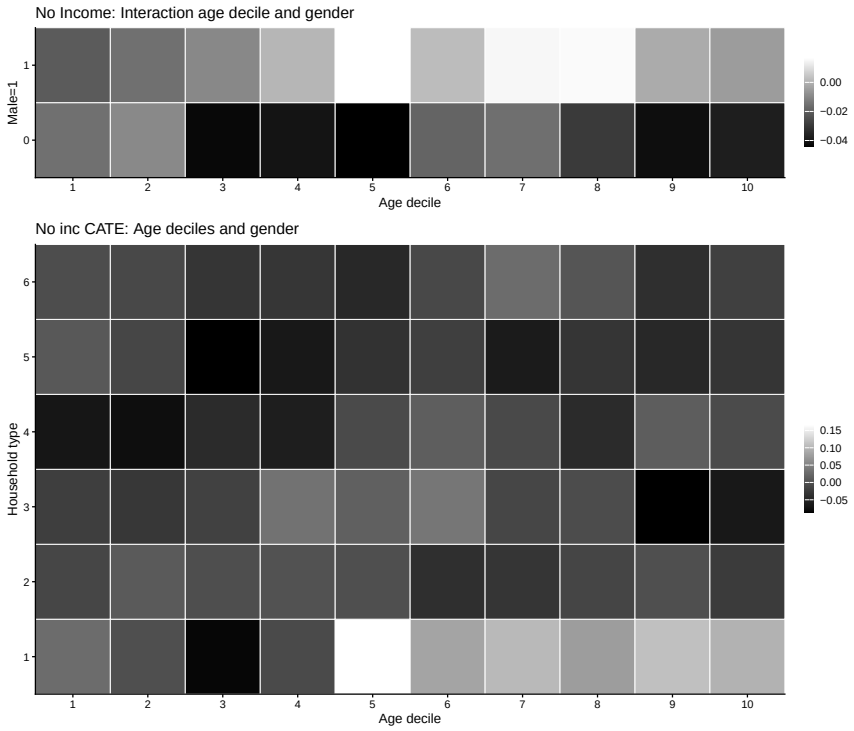


Note: CATE on having no registered income by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

On having no registered income, the mean estimated CATE is higher in magnitude for women, and especially non-single households. The estimated CATE is similar across migration backgrounds, except first generation non-western migrants which seem to response more. There is no clear pattern for age, which is surprising given that younger individuals tend to have higher labour market attachments, and have a higher chance to find a job. On the other hand, Markussen and Røed (2016) show that older and hard-to-employ individuals react to activation programmes by taking low-paying jobs, suggesting it may not be so surprising that older individuals respond to the RW by reducing their attachment to the informal or unpaid sector. In the context of the housing incentives provided by the policy, it may be that older individuals with strong idiosyncratic preferences to live in targeted neighbourhoods enter low-wage employment to ensure they can keep on living at their current address.

Figure A24 shows the interactions between CATEs. The upper panel shows the interaction between age decile and gender, and the lower panel the interaction between age decile and household types. Darker colours indicate larger magnitude effects (more negative in this case). The effects are clearly driven by women, non single households, and even older women. The strong effects found for the last age decile are unexpected, although not out of the norm given previous results in the literature.

Figure 15: **CATE on no income by pairs of characteristics - extensive margin**



Note: CATE on welfare participation by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

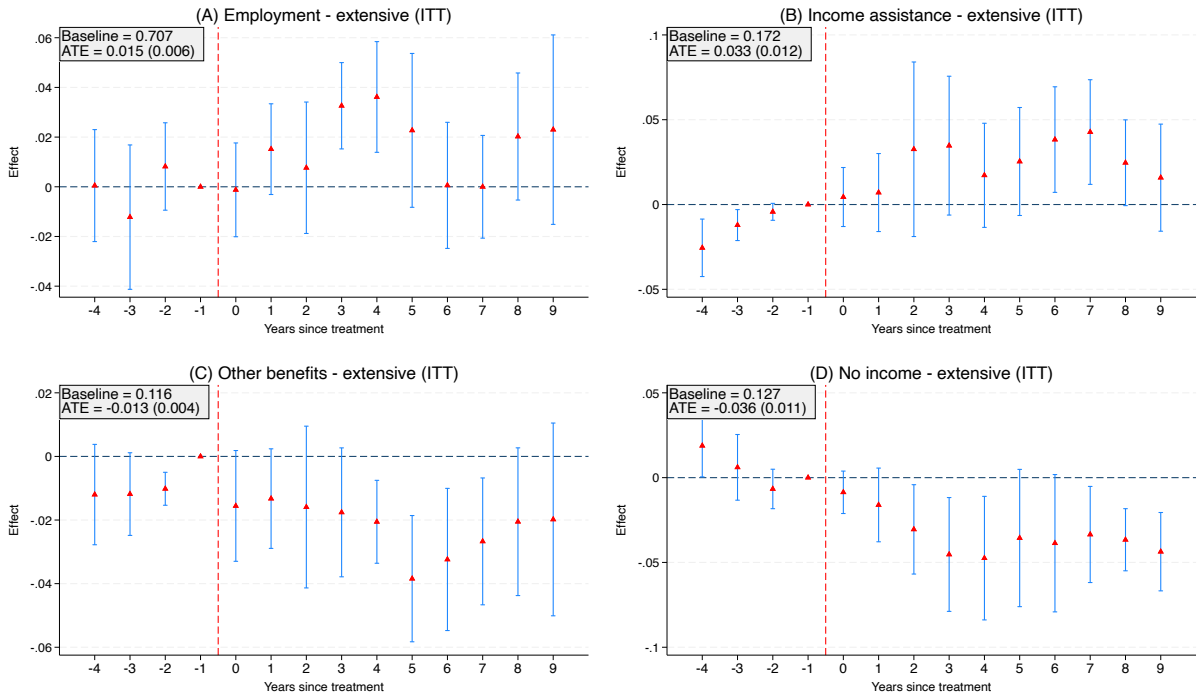
Figures A23, A24, A21, and A22 and show the results of running the causal forests using and other benefits as outcomes.

6.6 Spillovers and Placebo

To investigate how the policy spilled over we drop all treated neighbourhoods from the sample, and re-run the analysis using direct spillover neighbourhoods as treated units. Spillover neighbourhoods are all situated around fully treated neighbourhoods, in the south of the city (see Map A29).

Figure A25 shows the results of re-running the analysis with individuals in spillover neighbourhoods coded as treated, excluding individuals in treated areas, and using individuals in remaining areas as controls. This robustness check yields imprecisely estimated positive effects on employment and negative effects on other benefits and informal income take-up. The sign of the point estimates is the same as in our main analysis: employment for incumbent residents in spillover areas increases, whereas unregistered incomes decrease.

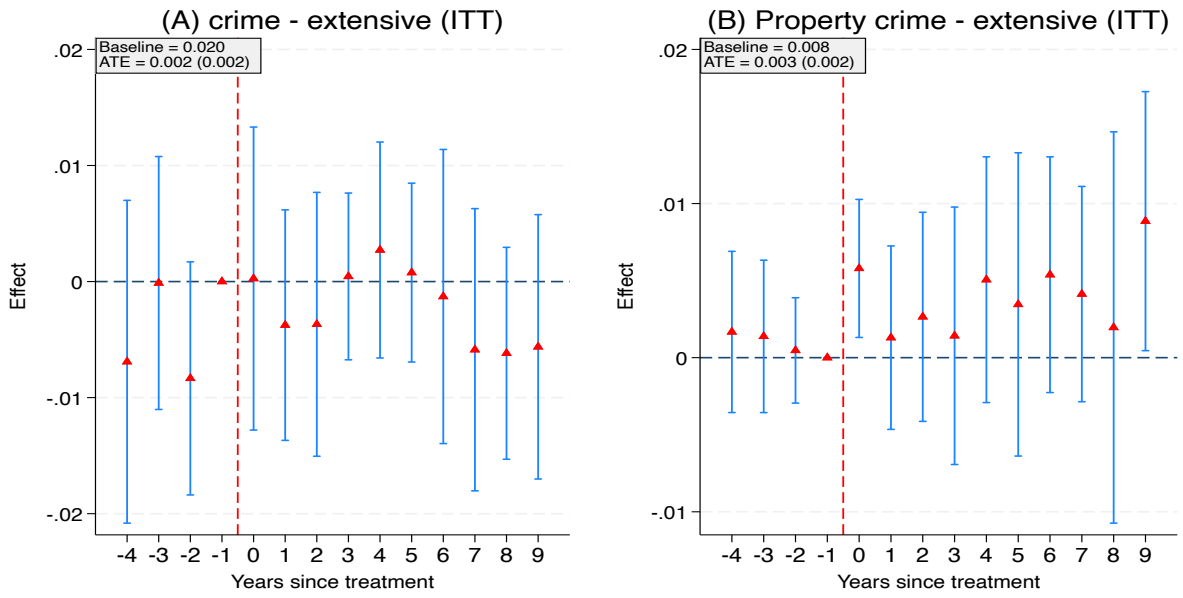
Figure 16: **Event study DDD on labour market outcomes for incumbents in spillover neighbourhoods**



Note: The plots show the results of estimating equation 2 on employment (panel A), welfare benefits (panel B), other benefits (panel C), and no registered income (panel D) at the extensive margin. All adult individuals living in spillover neighbourhoods are treated. All adults living in remaining untreated neighbourhoods (non-treated) are the controls.

The results of the spillover exercise suggest is related to the proximity of spillover neighbourhoods with treated neighbourhoods directly surround treated neighbourhoods (see Map A29). Residents in neighbouring locations could have also felt threatened by such a policy, such that a treatment effect is detectable in adjacent neighbourhoods. The lower magnitude of the results in Figure **??** is consistent with a treatment effect decaying farther away from treated areas. If this hypothesis is correct, our main specification underestimates the treatment effect.

Figure 17: **Event study DiD on labour market outcomes for incumbents in spillover neighbourhoods**



Note: The plots show the results of estimating equation 2 on all crimes (panel A) and property crimes (panel B). All adult individuals aged 20 to 45 living in spillover neighbourhoods are treated.

As a robustness check, we run a DDD specification on the same outcomes as in the main analysis, randomly choosing neighbourhoods in the RH region. Reassuringly, the results in Figures A27 and A28 show null effects, as would be expected absent any policy.

7 Conclusion

We study how neighbourhood composition affects incumbent residents using the Rotterdamwet, a policy allowing local governments to ban workless individuals from moving into specified neighbourhoods. Via selective in-migration, the policy increased the share of residents in employment, whilst reducing the share of residents on welfare and the share of criminal offenders. We document a behavioural response for incumbent residents, who react by increasing employment and reducing informal or unpaid work take-up. We document a reduction in incumbents' probability to commit a crime. The effects on employment and welfare receipt persist 4 to 5 years after implementation, but fade out afterwards. Analysing heterogeneity using causal forests reveals that the increase in employment for incumbents is driven by women and younf individuals ages 20 to 35. The effects on unpaid/undeclared income are driven by non-single households and women. We find no effects of the policy on youth outcomes, consistent with early analyses of the MTO experiment. Exploiting a unique of the feature of the Rotterdamwet, which only applies to individuals with less than 6 years of residency tenure in the region we detect a strong response to the threat of displacement for incumbents. We cannot rule out that the changes in composition had effects through peers. Future analyses leveraging detailed dwelling-level data will establish whether this is indeed the case.

Our results show that incentives through housing displacement and possibly neighbourhood composition

affects adult labour market outcomes, which contrasts with earlier findings in the literature. The effects on formal work take-up we and crime could be driven by changes in peer composition, a possible channel mediating place effects for adult incumbent residents. Moreover, we show that incentives delivered through housing mobility restrictions can act as a break in welfare cultures. To keep on living in targeted neighbourhoods, residents have incentives to find a job and continue working in the medium-term, with non-trivial effects on welfare dependence and employment.

Our analysis doesn't explore the policy's impact on individuals who would have lived in targeted neighbourhoods absent the policy. The costs borne by excluded individuals could potentially outweigh the positive effects we identify for long-term residents. Future analyses examining the outcomes of excluded individuals could shed light on this question.

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A Treated streets, neighbourhood selection, and additional maps

Table A1: Treated Streets, Corresponding Neighbourhoods, and Year Treated

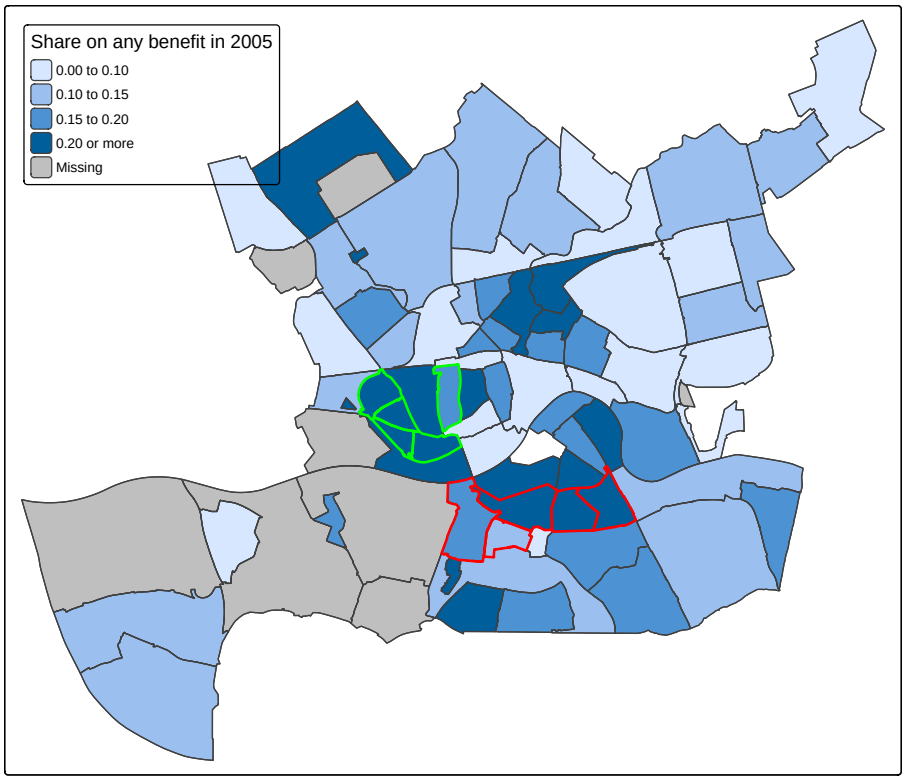
Street	**Neighborhood**	**Year**
Groene Hilledijk (above Strevelseweg)	Bloemhof	2006
Putsebocht	Bloemhof	2006
Strevelsweg (only in Bloemhof)	Bloemhof	2006
Dordtselaan	Bloemhof	2006
Hillevliet	Bloemhof	2006
Slaghekstraat	Bloemhof	2006
Riederstraat	Bloemhof	2006
Moerkerksestraat	Bloemhof	2006
Borselaarstraat	Bloemhof	2006
Bas Jungeriusstraat	Bloemhof	2006
Katendrechtse Lagedijk	Bloemhof	2006
Mathenesserdijk	Tussendijken	2006
Dirk Danestraat	Spangen	2006
Wallisweg	Spangen	2006
Vosmaerstraat	Spangen	2006
Mathenesserweg	Tussendijken/Spangen	2006
Grote Visserijstraat	Tussendijken	2006
1^{e} Middellandstraat	Middelland	2006
Middellandplein	Middelland	2014
Schiedamseweg	Delfshaven/Bospolder	2014
's Gravendijkwal	Delfshaven	2014
Grote Visserijstraat	Tussendijken	2018
Mathenesserdijk	Tussendijken	2018
Mathenesserweg	Tussendijken/Spangen	2018

Note: for streets crossing multiple neighbourhoods, I only mention the section of the streets that were treated.

Variable	Relation with Problem Cumulation Score
% single-person households	-
% Housing vacancies	+
% Long-term unemployment	-
% Overcrowding	+
% Non-Western immigrants	+
% Approved rental housing	+
% Share newcomers	+
% Welfare recipients	+
% Mobility	+
% Housing (WOZ) - value per room	-
% Reported neighborhood nuisance	+
% Reported noise pollution	+

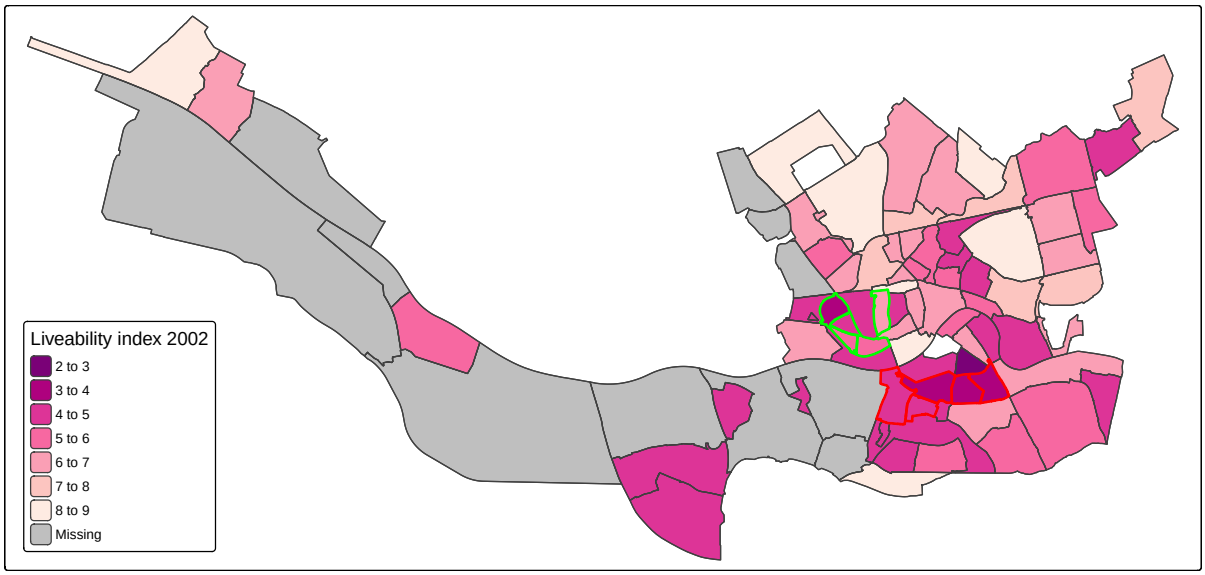
Table A2: Criteria used to calculate the 'problem cumulation score' used to select treated neighbourhoods.

Figure A1: **Treated areas and the spatial variation in government benefits reliance in Rotterdam (2005)**



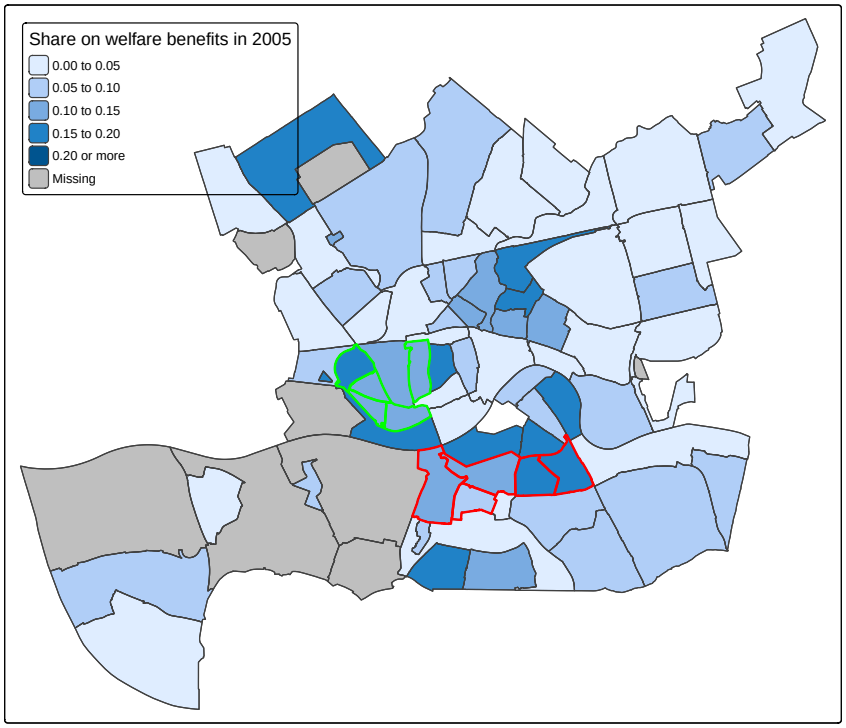
Note: The borders of **fully treated** neighbourhoods (as of 2006) are represented in **red**, whereas **partially treated** neighbourhoods (as of 2014) are in **green**. The industrial terrains West of the city are excluded for clarity.

Figure A2: **Leefbarometer (liveability) index by neighbourhood in Rotterdam**



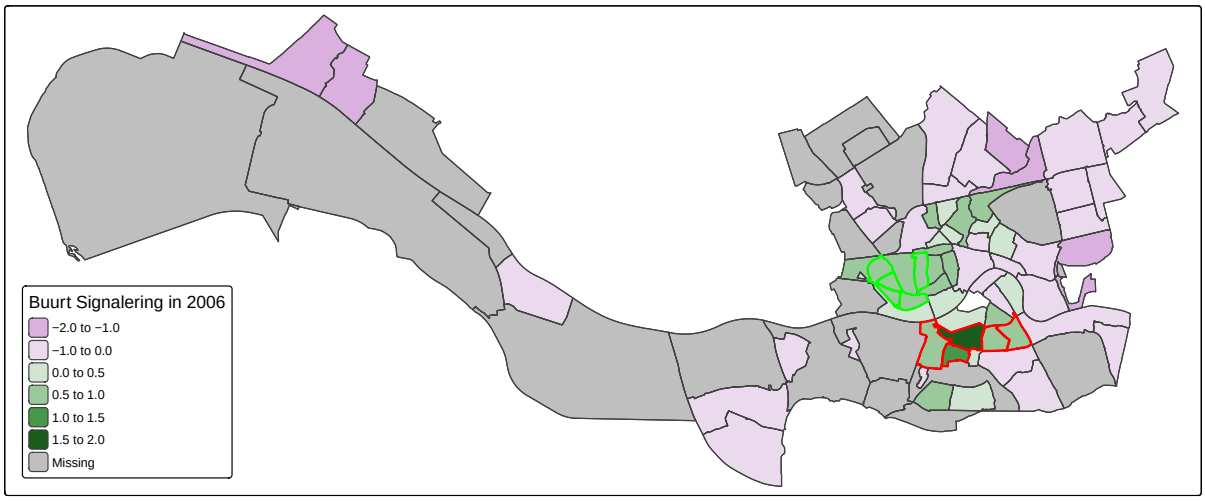
Note: The borders of fully treated neighbourhoods (as of 2006) are represented in red, whereas partially treated neighbourhoods (as of 2014) are in green.

Figure A3: **Share of individuals on income assistance by neighbourhood in Rotterdam.**



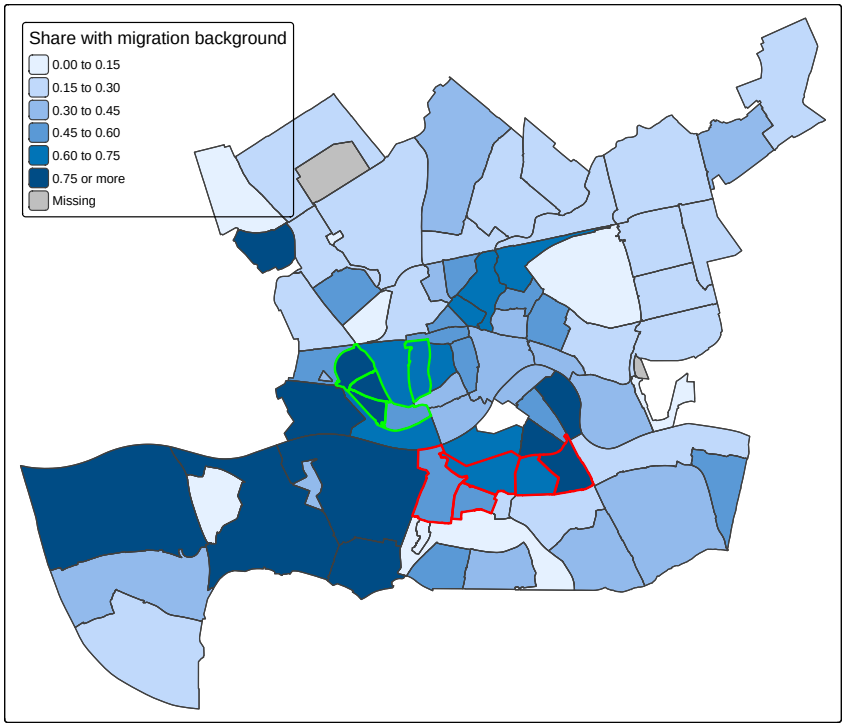
Note: The borders of fully treated neighbourhoods (as of 2006) are represented in red, whereas partially treated neighbourhoods (as of 2014) are in green. The industrial terrains West of the city are excluded for clarity.

Figure A4: **Buurtsignalering (neighbourhood alert) index by neighbourhood in Rotterdam**



Note: The borders of fully treated neighbourhoods (as of 2006) are represented in red, whereas partially treated neighbourhoods (as of 2014) are in green.

Figure A5: **Share of individuals with a migration background by neighbourhood in Rotterdam**



Note: The borders of fully treated neighbourhoods (as of 2006) are represented in red, whereas partially treated neighbourhoods (as of 2014) are in green. The industrial terrains West of the city are excluded for clarity.

B Additional Summary statistics

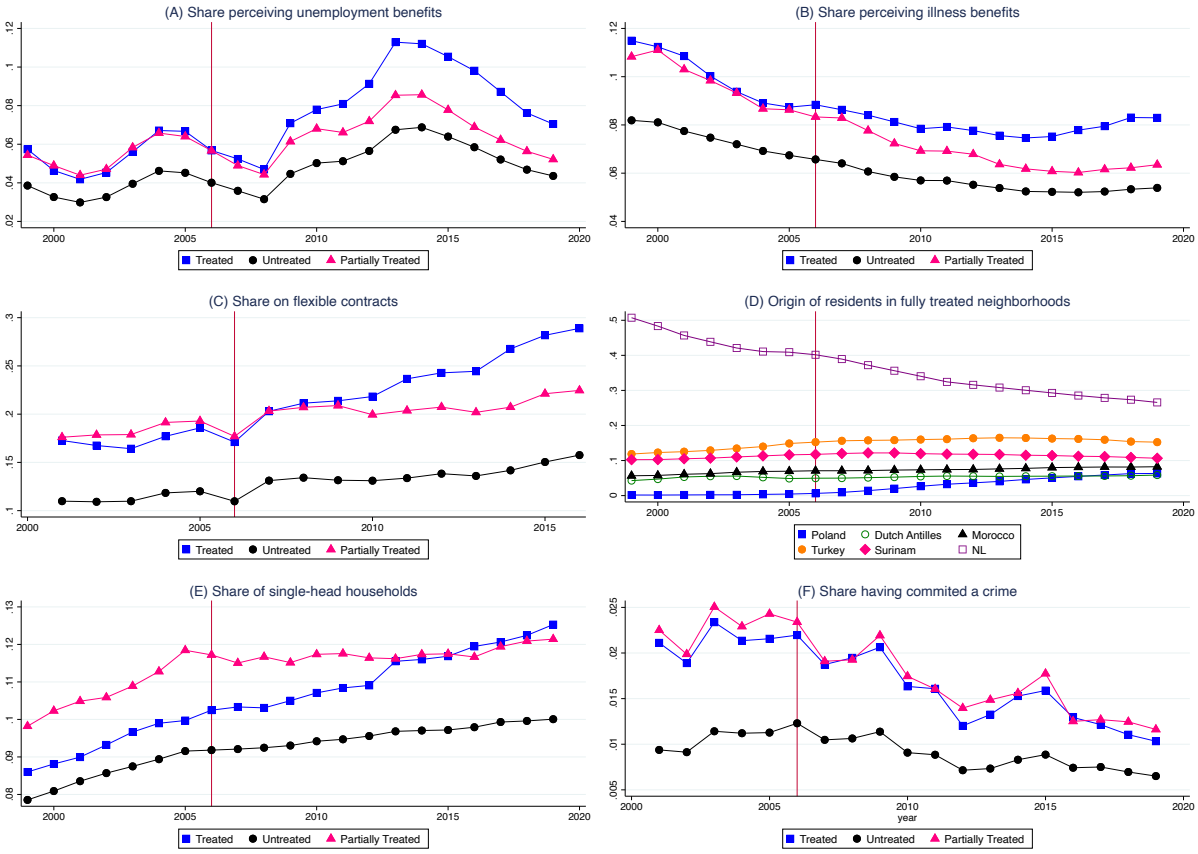
B.1 Composition

Table A3: Descriptive Statistics for the WOZ Unrestricted Sample

	Treated	Control	Norm. Diff	P-val
Age	38.280	39.590	−0.104	0.058
Male	0.530	0.510	0.042	0.001
Single Head HH	0.100	0.110	−0.036	0.122
Turkish	0.160	0.110	0.161	0.171
Moroccan	0.070	0.070	0.016	0.717
Antillian	0.050	0.040	0.078	0.030
Surinamese	0.120	0.120	0.013	0.733
Native	0.370	0.460	−0.181	0.127
First Generation	0.520	0.430	0.165	0.125
Second Generation	0.110	0.110	0.021	0.356
Criminal Record	0.020	0.020	0.021	0.104
Employed	0.600	0.620	−0.037	0.546
Welfare	0.190	0.150	0.100	0.100
Unemployment Ins.	0.080	0.060	0.053	0.000
Sickness Benefits	0.100	0.110	−0.016	0.437
Observations	36,714	304,677		

Notes: This table presents the means of baseline characteristics for the treated and control groups for individuals aged 20-65 in the house price (WOZ) quartile trim. The normalized difference (Imbens and Wooldridge, 2009) and the p-value of the difference in means are reported in the final two columns of each panel. Standard errors are clustered at the neighbourhood (*buurt*) level.

Figure A6: **Additional descriptives - composition by treatment status**



Note: eligibility to unemployment and sickness benefits is conditional on having worked 26 weeks out of 36 before unemployment. Flexible contracts are on-call or temp agency contracts. Crime is measured as the share of residents having committed a crime in a given year.

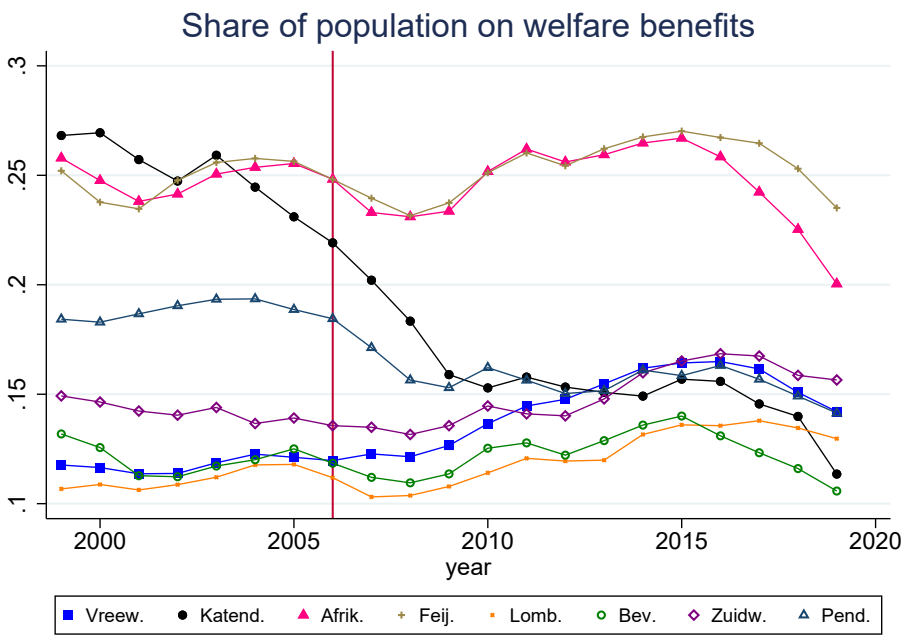


Figure A7: Share of residents on welfare benefits in selected neighbourhoods of Rotterdam.

Table A4: **Descriptive statistics on individuals in treated and control neighbourhoods in 2005 (Incumbents)**

	Full Sample				**KNN Unrestricted**			
	Treated	Control	Norm. Diff	P-val	Treated	Control	Norm. Diff	P-val
Age	39.95	42.82	−0.236	0.000	39.95	40.72	−0.063	0.259
Male	0.51	0.50	0.040	0.002	0.51	0.50	0.025	0.081
Single head hh	0.11	0.08	0.083	0.001	0.11	0.12	−0.043	0.111
Turkish	0.17	0.05	0.399	0.003	0.17	0.09	0.230	0.074
Moroccan	0.07	0.03	0.194	0.000	0.07	0.08	−0.012	0.799
Antillian	0.04	0.02	0.137	0.000	0.04	0.03	0.050	0.147
Surinamese	0.12	0.07	0.191	0.000	0.12	0.14	−0.042	0.301
Native	0.39	0.66	−0.575	0.000	0.39	0.44	−0.103	0.409
First Generation	0.51	0.25	0.566	0.000	0.51	0.46	0.107	0.359
Second Generation	0.10	0.09	0.041	0.040	0.10	0.10	−0.010	0.627
Criminal Record	0.02	0.01	0.068	0.000	0.02	0.02	0.008	0.535
Employed	0.59	0.67	−0.160	0.009	0.59	0.61	−0.034	0.597
Welfare	0.19	0.08	0.320	0.000	0.19	0.15	0.084	0.172
Unemployment Insurance	0.08	0.05	0.113	0.000	0.08	0.06	0.062	0.000
Sickness Benefits	0.11	0.09	0.067	0.002	0.11	0.11	−0.006	0.807
Observations (N)	30,317	781,677			30,317	204,023		

Notes: This table presents the means of baseline characteristics for the treated and control groups across the Full Sample and the KNN sample for incumbents. The normalized difference and the p-value of the difference in means are reported in the final two columns of each panel. Standard errors are clustered at the neighbourhood (*buurt*) level.

Table A5: **Descriptive statistics on incumbent individuals in treated and control neighbourhoods in 2005 (WOZ Unrestricted)**

	Treated	Control	Norm. Diff	P-val
Age	39.95	41.17	−0.099	0.078
Male	0.51	0.50	0.035	0.014
Single head hh	0.11	0.12	−0.035	0.180
Turkish	0.17	0.11	0.183	0.141
Moroccan	0.07	0.07	0.022	0.647
Antillian	0.04	0.03	0.054	0.110
Surinamese	0.12	0.12	0.013	0.745
Native	0.39	0.48	−0.177	0.160
First Generation	0.51	0.43	0.170	0.150
Second Generation	0.10	0.10	0.009	0.658
Criminal Record	0.02	0.02	0.007	0.564
Employed	0.59	0.61	−0.043	0.498
Welfare	0.19	0.15	0.101	0.104
Unemployment Insurance	0.08	0.07	0.055	0.000
Sickness Benefits	0.11	0.11	−0.014	0.519
Observations (N)	30,317	256,204		

Notes: This table presents the means of baseline characteristics for the treated and control groups within the WOZ Unrestricted sample for incumbents. The normalized difference and the p-value of the difference in means are reported in the final two columns. Standard errors are clustered at the neighbourhood (*buurt*) level.

Table A6: **Descriptive statistics on individuals in treated and control neighbourhoods in 2005 (Incumbents, KNN excluding untreated in Rotterdam)**

	Treated	Control	Norm. Diff	P-val
Age	39.95	40.58	−0.052	0.360
Male	0.51	0.51	0.013	0.395
Single head hh	0.11	0.12	−0.033	0.293
Turkish	0.17	0.10	0.200	0.135
Moroccan	0.07	0.08	−0.026	0.657
Antillian	0.04	0.03	0.039	0.257
Surinamese	0.12	0.15	−0.077	0.118
Native	0.39	0.41	−0.054	0.690
First Generation	0.51	0.49	0.045	0.724
Second Generation	0.10	0.10	0.013	0.562
Criminal Record	0.02	0.02	−0.006	0.619
Employed	0.59	0.60	−0.013	0.851
Welfare	0.19	0.15	0.107	0.115
Unemployment Insurance	0.08	0.06	0.068	0.000
Sickness Benefits	0.11	0.12	−0.038	0.128
Observations (N)	30,317	121,285		

Notes: This table presents the means of baseline characteristics for the treated and control groups within the KNN sample for incumbents (excluding the untreated in Rotterdam). The normalized difference and the p-value of the difference in means are reported in the final two columns. Standard errors are clustered at the neighbourhood (*buurt*) level.

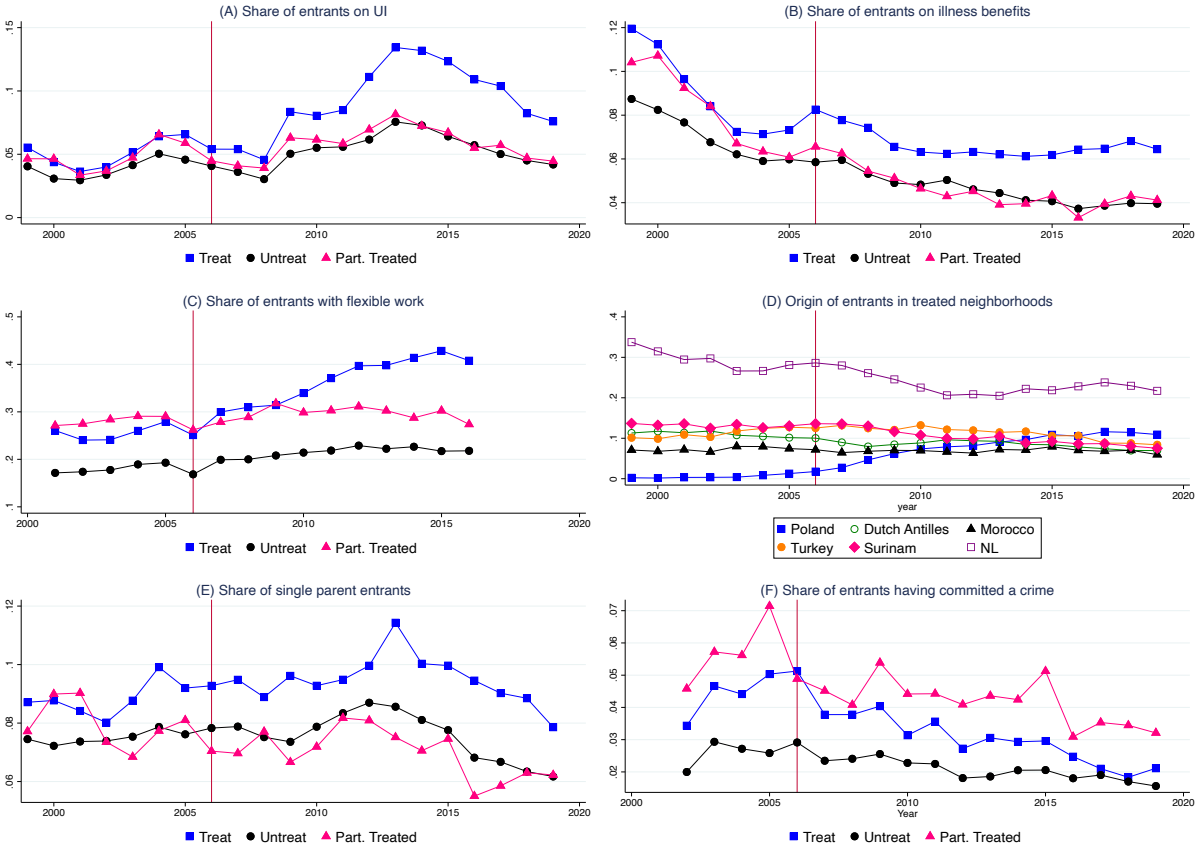
Table A7: **Table of summary statistics with the characteristics of residents receiving welfare benefits in treated and control neighbourhoods in 2005**

	Group	
	Treated	Control
Age	41.40	43.60
Turkish Origin	0.20	0.11
Moroccan Origin	0.12	0.13
Antillian Origin	0.11	0.08
Surinam Origin	0.14	0.15
Native	0.21	0.30
1st Gen. Migrant	0.70	0.60
2nd Gen. Migrant	0.09	0.10
Single Parent	0.23	0.28
Crime	0.05	0.04
Welfare ben.	1.00	1.00
Unemployment ben.	0.04	0.04
Disability ben.	0.04	0.04

Note: Summary statistics conditional on perceiving welfare benefits in 2005.

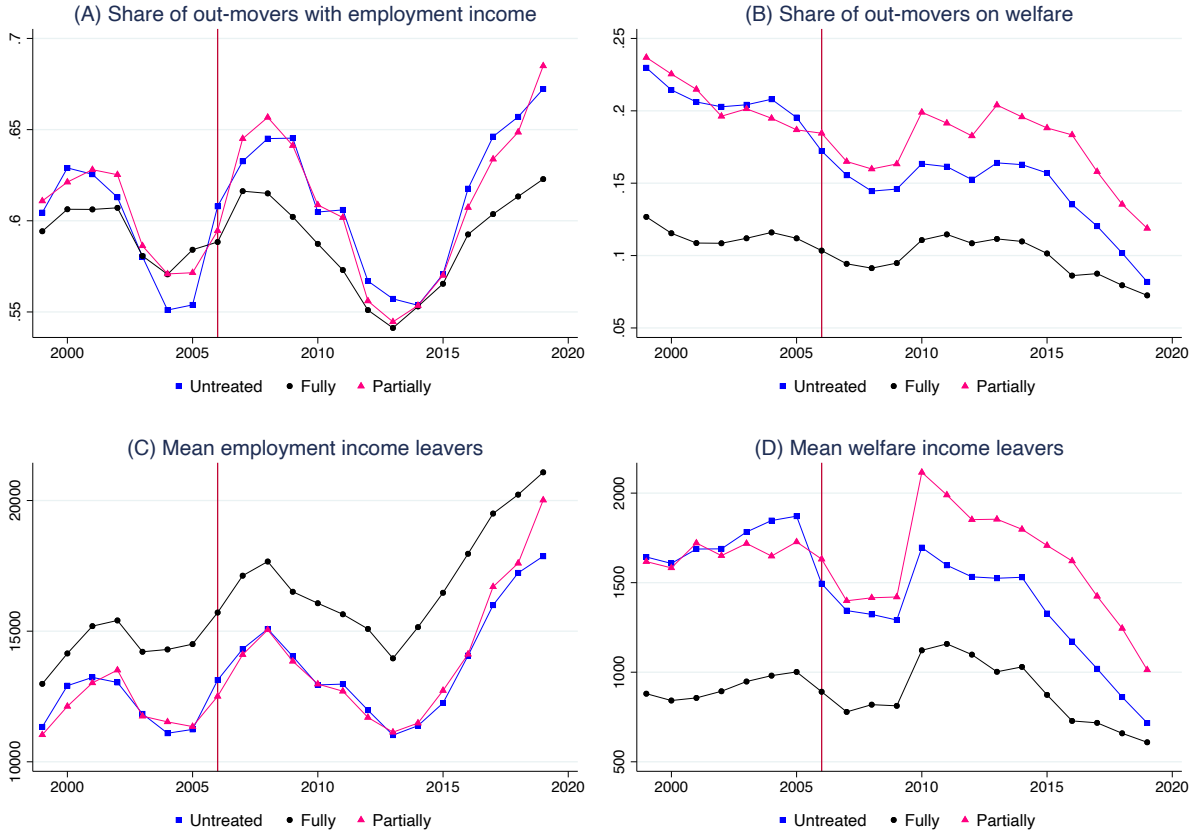
Entrants and leavers

Figure A8: **Additional descriptives on entrants by treatment status**



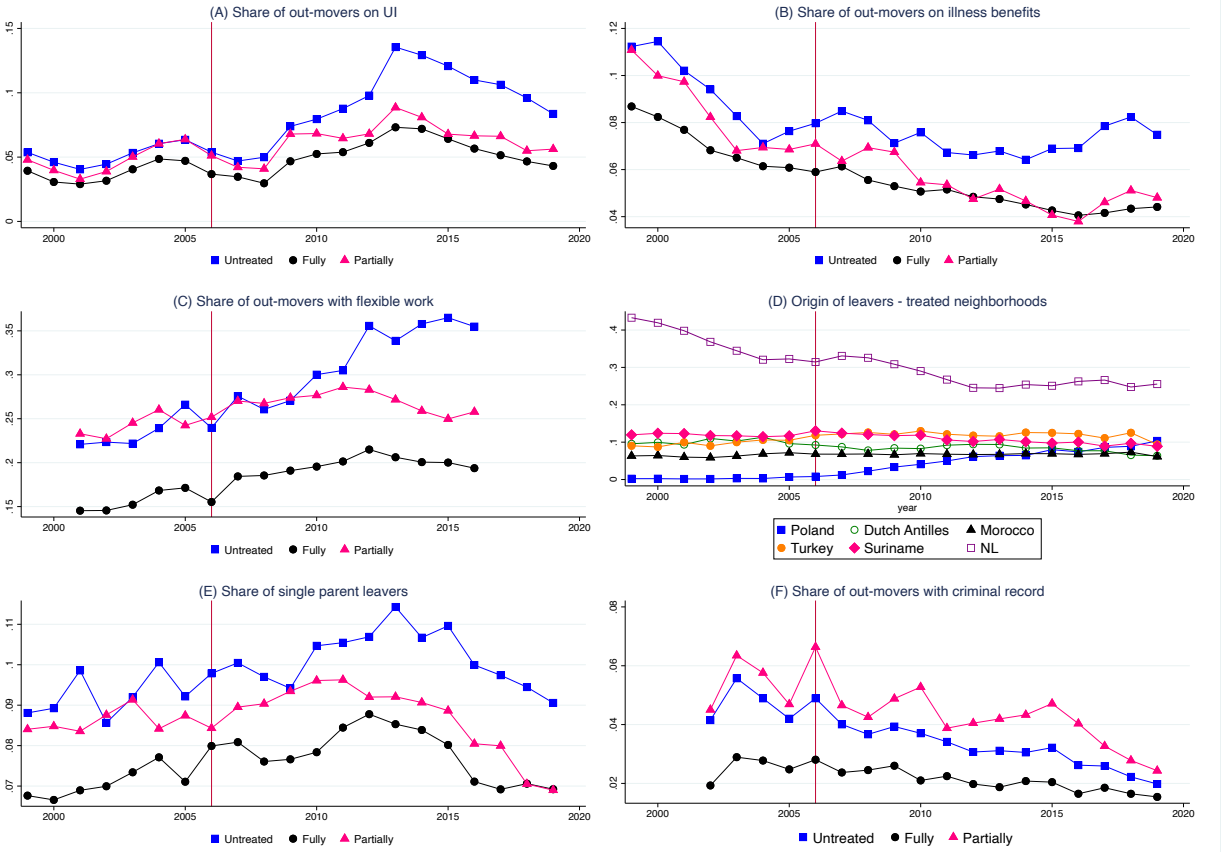
Note: eligibility to UI and illness benefits are conditional on 26 out of the last 36 weeks before unemployment. We code on-call and temp-agency contracts as flexible. Crime is calculated as the share having committed a criminal offence in the last year.

Figure A9: **Descriptives on leavers by treatment status**



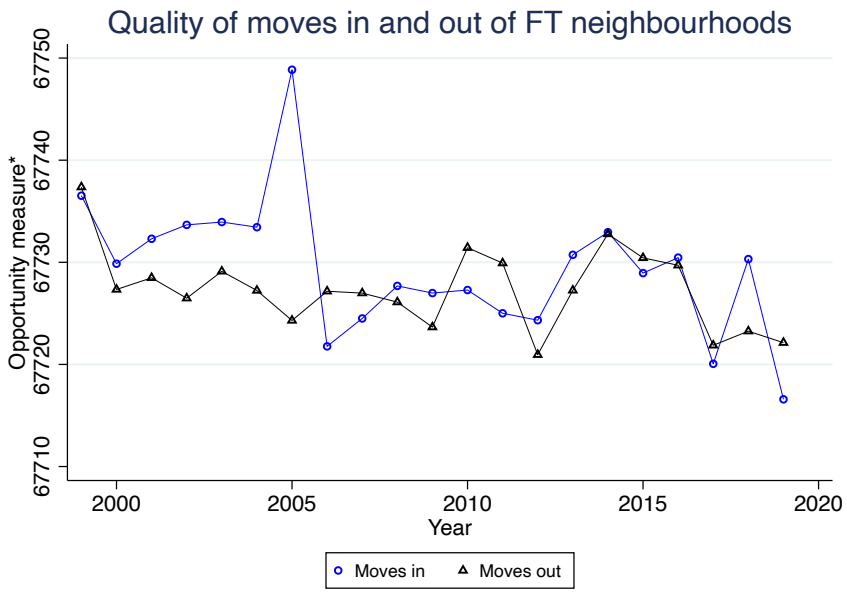
Note: Other benefits include UI, disability and sickness benefits.

Figure A10: **Additional descriptives on leavers by treatment status**



Note: eligibility to UI and illness benefits are conditional on 26 out of the last 36 weeks before unemployment. We code on-call and temp-agency contracts as flexible. Crime is calculated as the share having committed a criminal offence in the last year.

Figure A11: **Intergenerational mobility in the neighbourhoods individuals in treated neighbourhoods move from, and neighbourhoods they move into**



Note: Figure A11 is based on data compiled by Lam et al. (2025). The plot shows the mean income of children at 30 (given their parents being between the 20th and 30th household income percentile) in neighbourhoods where individuals residing in fully treated neighbourhoods move from/move into.

B.2 Triple diff balancing table - unrestricted and incumbents

Table A8: **Descriptive statistics on individuals in treated and control neighbourhoods (± 6 years, Full Sample, age 25-45)**

	Treated		Control			
	- 6 years	+ 6 years	- 6 years	+ 6 years	P-val DiD	Norm. DiD
Age	29.48	32.75	30.55	34.02	0.356	−0.030
Male	0.54	0.51	0.50	0.50	0.235	−0.050
Single head hh	0.09	0.12	0.07	0.10	0.486	0.019
Turkish	0.10	0.22	0.05	0.06	0.000	0.437
Moroccan	0.07	0.09	0.04	0.03	0.000	0.101
Antillian	0.10	0.04	0.05	0.02	0.000	−0.121
Surinamese	0.11	0.14	0.06	0.06	0.004	0.103
Native	0.27	0.34	0.50	0.68	0.000	−0.243
First Generation	0.64	0.49	0.40	0.19	0.019	0.114
Second Generation	0.09	0.18	0.10	0.12	0.000	0.203
Criminal Record	0.04	0.02	0.02	0.01	0.211	−0.043
Employed	0.64	0.69	0.72	0.79	0.245	−0.036
Welfare	0.19	0.18	0.10	0.08	0.364	0.034
Unemployment Insurance	0.08	0.09	0.05	0.05	0.004	0.036
Sickness Benefits	0.06	0.08	0.04	0.05	0.001	0.048
Observations (N)	9,000	7,015	81,555	272,921		

Notes: This table presents the means of baseline characteristics for the treated and control groups comparing the period −6 years and +6 years around the intervention. The sample includes all individuals aged 25-45 in the **Full Sample**. The p-value of the difference-in-differences (DiD) and the normalized DiD are reported in the final two columns. Standard errors are clustered at the neighbourhood (*buurt*) level.

Table A9: **Descriptive statistics on individuals in treated and control neighbourhoods (± 6 years, Full Sample Incumbents, age 25-45)**

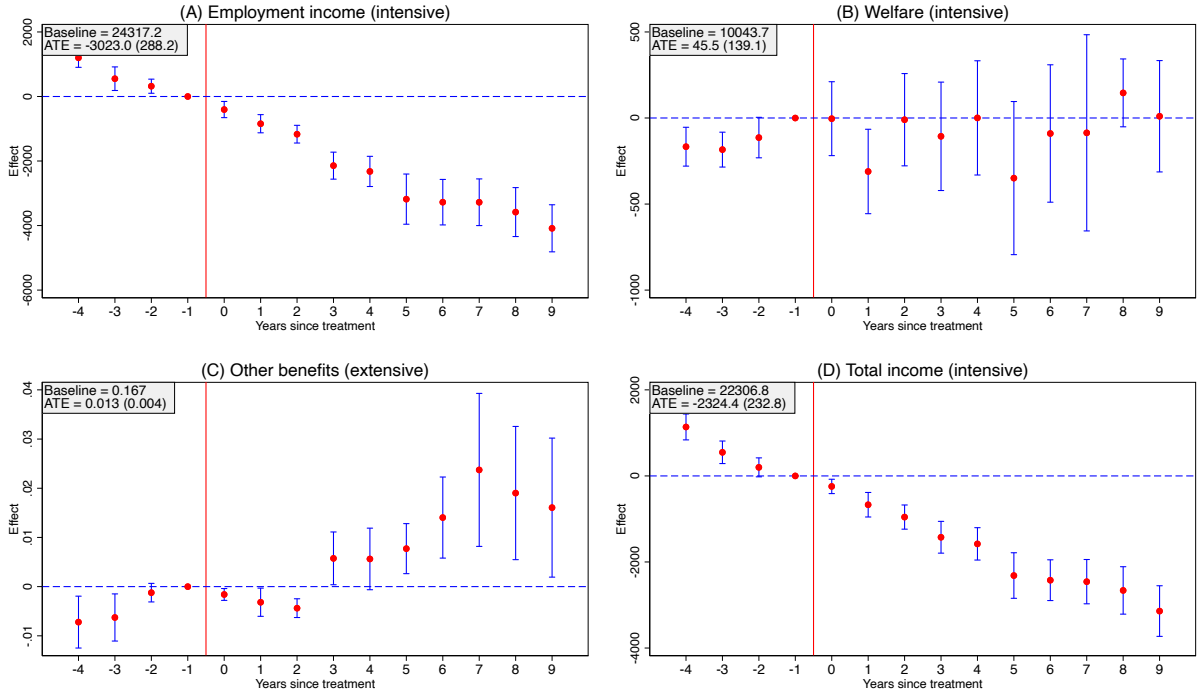
	Treated		**Control**			
	- 6 years	+ 6 years	- 6 years	+ 6 years	P-val DiD	Norm. DiD
Age	30.09	33.69	31.39	34.82	0.509	0.026
Male	0.53	0.51	0.50	0.49	0.376	−0.032
Single head hh	0.09	0.13	0.07	0.10	0.299	0.025
Turkish	0.11	0.23	0.05	0.06	0.000	0.421
Moroccan	0.08	0.09	0.04	0.03	0.001	0.074
Antillian	0.08	0.03	0.05	0.02	0.000	−0.083
Surinamese	0.11	0.14	0.06	0.06	0.006	0.094
Native	0.28	0.34	0.51	0.69	0.000	−0.249
First Generation	0.64	0.51	0.40	0.20	0.004	0.138
Second Generation	0.08	0.15	0.09	0.11	0.000	0.180
Criminal Record	0.03	0.02	0.02	0.01	0.326	−0.032
Employed	0.64	0.68	0.73	0.78	0.190	−0.045
Welfare	0.17	0.18	0.10	0.08	0.044	0.082
Unemployment Insurance	0.08	0.09	0.05	0.05	0.049	0.036
Sickness Benefits	0.06	0.08	0.04	0.05	0.001	0.055
Observations (N)	5,647	4,629	56,183	239,986		

Notes: This table presents the means of baseline characteristics for the treated and control groups comparing the period −6 years and +6 years around the intervention. The sample includes **Incumbents** aged 25-45 in the Full Sample. The p-value of the difference-in-differences (DiD) and the normalized DiD are reported in the final two columns. Standard errors are clustered at the neighbourhood (*buurt*) level.

C Additional Figures - DiD

C.1 Event study DiD on composition - intensive margin

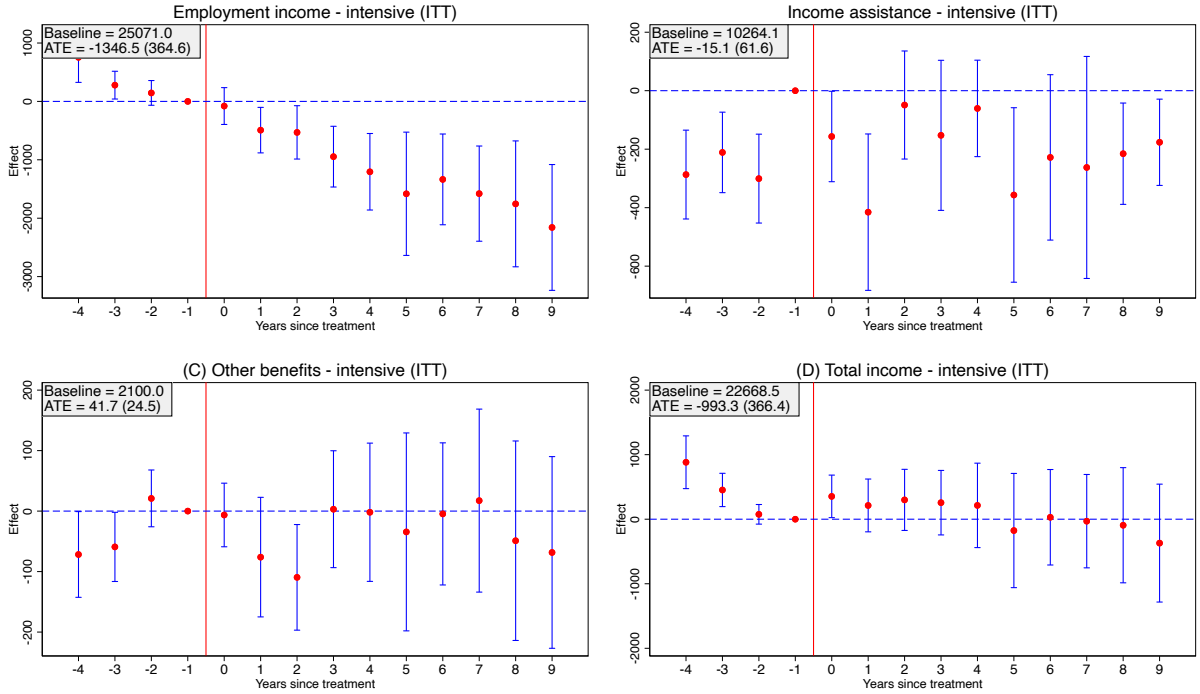
Figure A12: **Event study on labour market outcomes - composition (intensive margin)**



Note: The plots show the results of estimating equation 1 on employment income (panel A), welfare benefits (panel B), illness/disability (panel C), and unemployment benefits (panel D) at the intensive margin. All adult individuals living in fully treated neighbourhoods are treated. All adults living in untreated neighbourhoods, excluding partially treated and Katendrecht are the controls.

C.2 Event study DiD on incumbents - intensive margin KNN

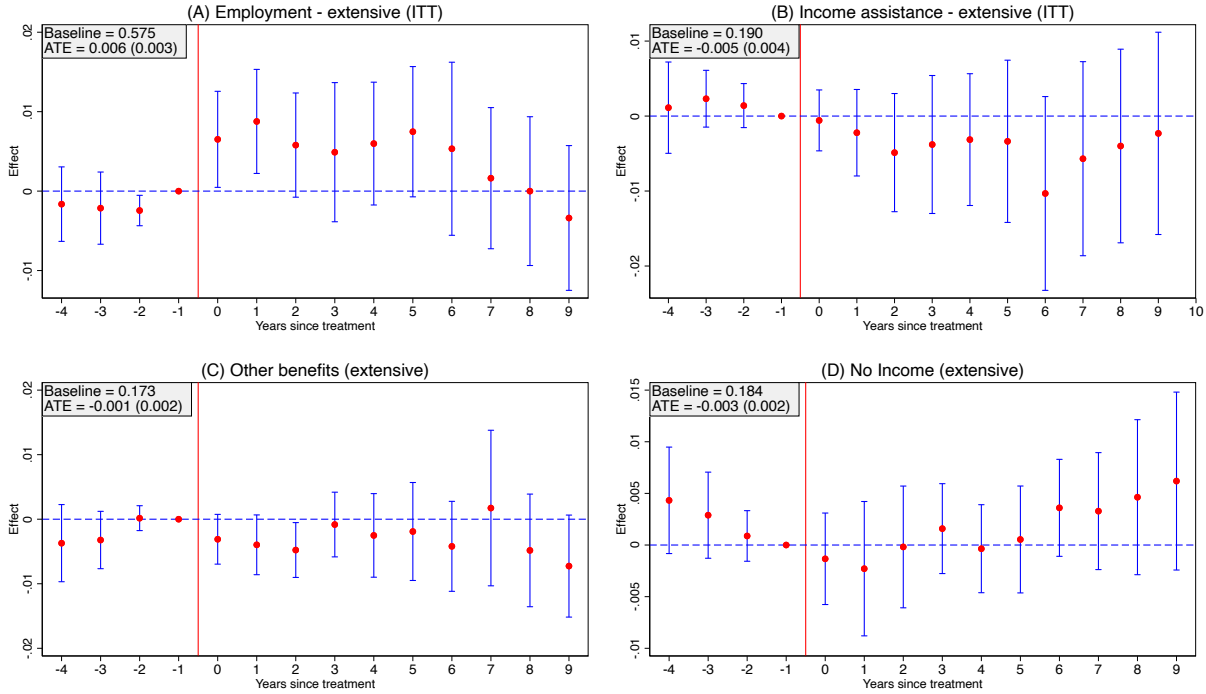
Figure A13: **Event study DiD on labour market outcomes - incumbents (intensive margin)**



Note: The plots show the results of estimating equation 1 on employment (panel A), welfare benefits (panel B), other benefits (panel C) at the extensive margin. Panel D shows the effects on total income (employment income and transfers). All adult individuals living in the same neighbourhood three years before treatment. We excluding individuals in partially treated neighbourhoods and Katendrecht from the analysis.

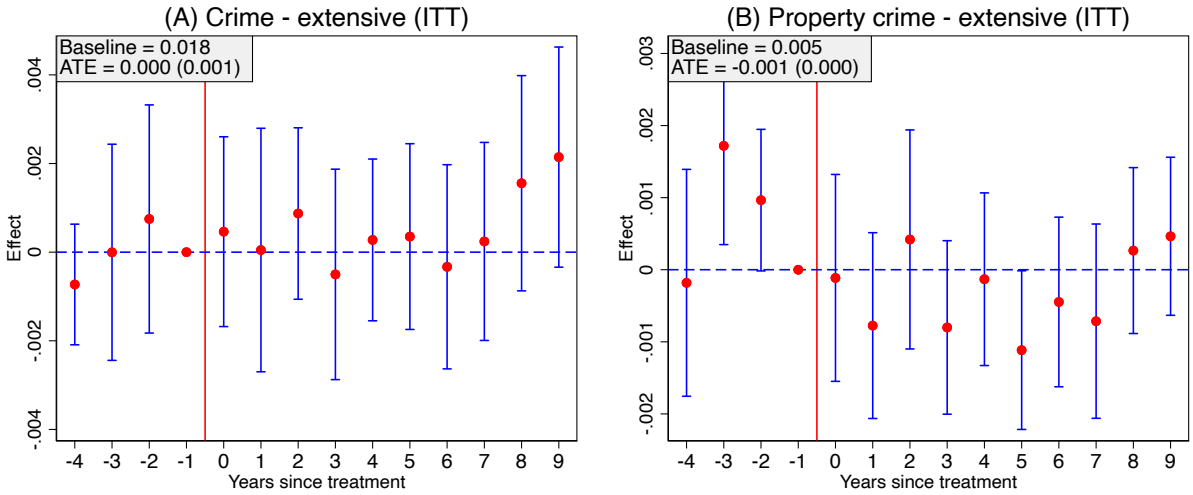
C.3 Event study DiD on incumbents - WOZ

Figure A14: **Event study DiD on incumbents' labour market outcomes - extensive margin**



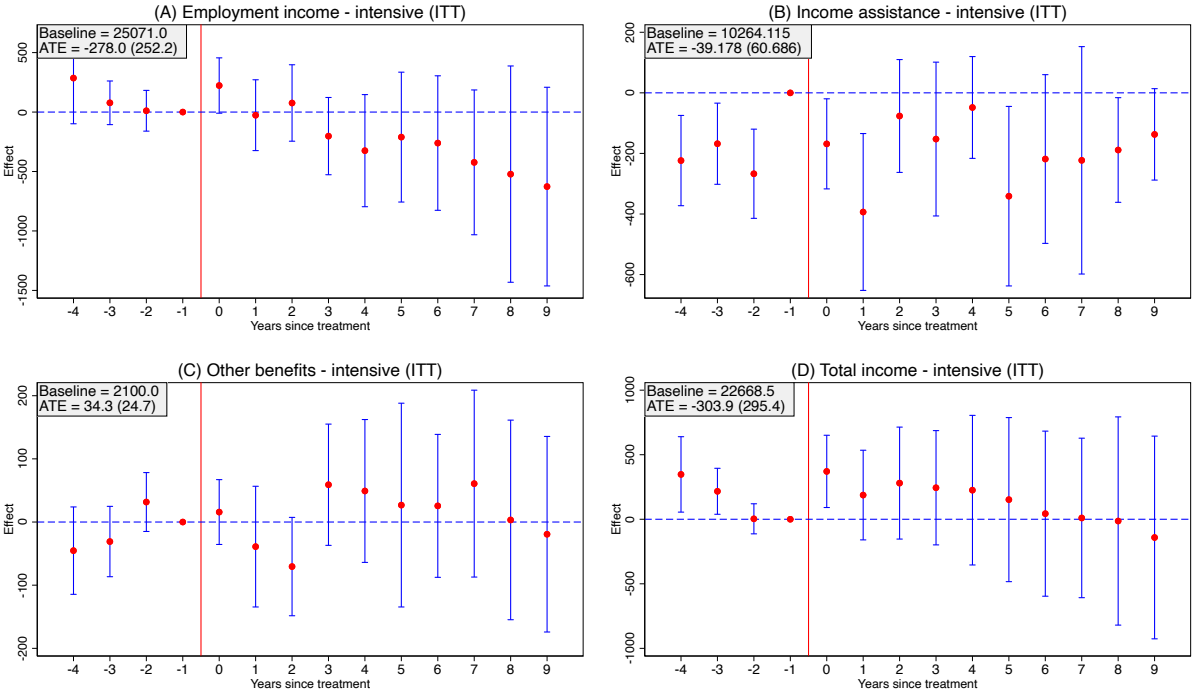
Note: The plots show the results of estimating Equation (1). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on total income at the intensive margin (including zeroes). The baselines are the 2005 variable mean for incumbents in *treated* neighbourhoods.

Figure A15: **Event study DiD on crime for incumbents (WOZ)**



Note: The plots show the results of estimating Equation (1). Panel A: effects on probability to commit a crime. The baseline is the 2005 variable mean for incumbents in *treated* neighbourhoods.

Figure A16: **Event study DiD on labour market outcomes - incumbents (intensive margin)**

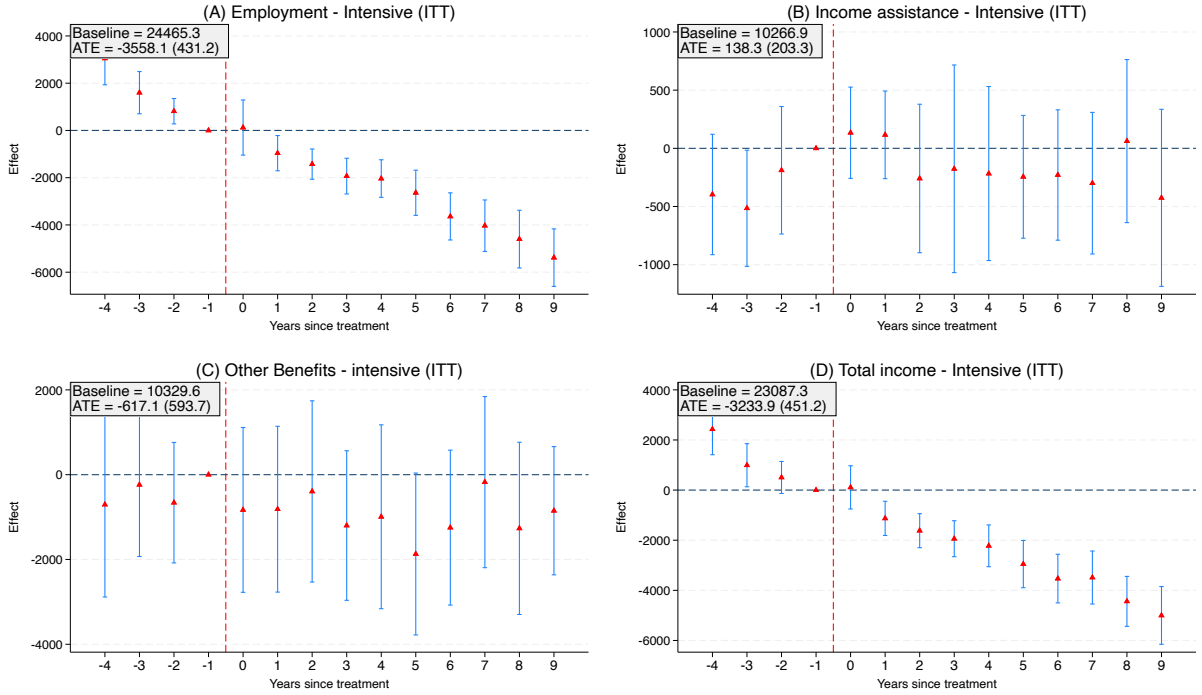


Note: The plots show the results of estimating equation 1 on employment (panel A), welfare benefits (panel B), other benefits (panel C) at the extensive margin. Panel D shows the effects on total income (employment income and transfers). All adult individuals living in the same neighbourhood three years before treatment. We excluding individuals in partially treated neighbourhoods and Katendrecht from the analysis.

D Additional Figures - DDD

D.1 Intensive margin DDD - KNN sample

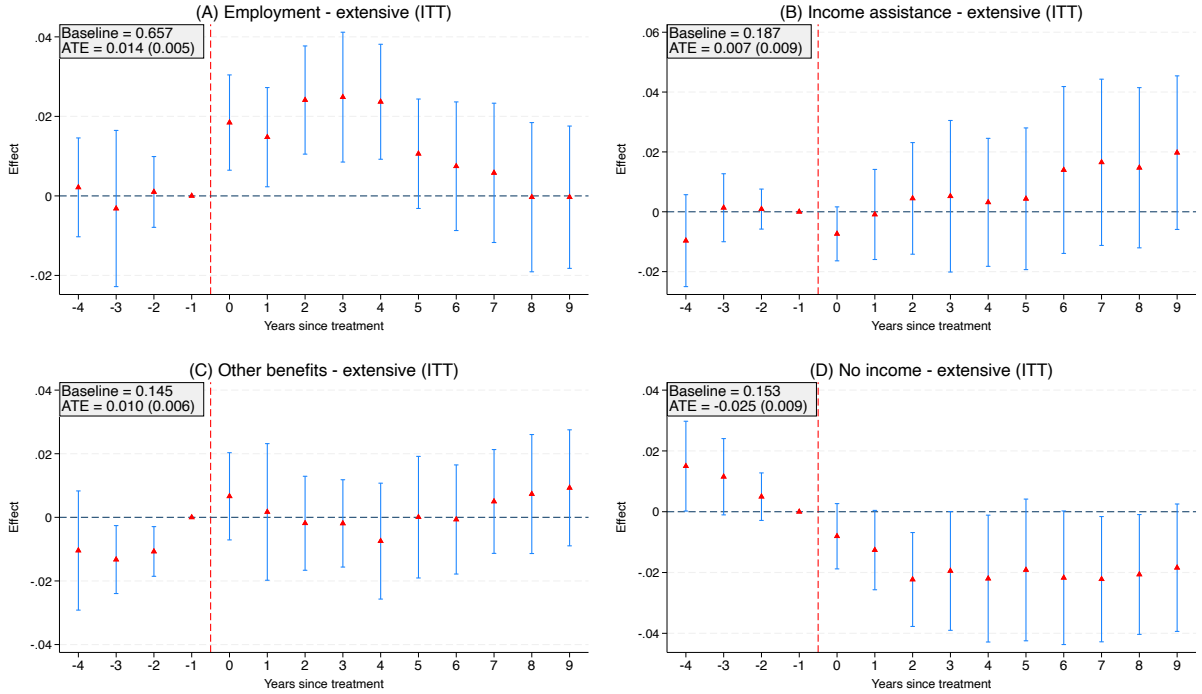
Figure A17: **Event study DDD on incumbents' labour market outcomes - extensive margin**



Note: The plots show the results of estimating Equation (2). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on total income at the intensive margin (including zeroes). The baselines are the 2005 variable mean for incumbents in *treated* neighbourhoods.

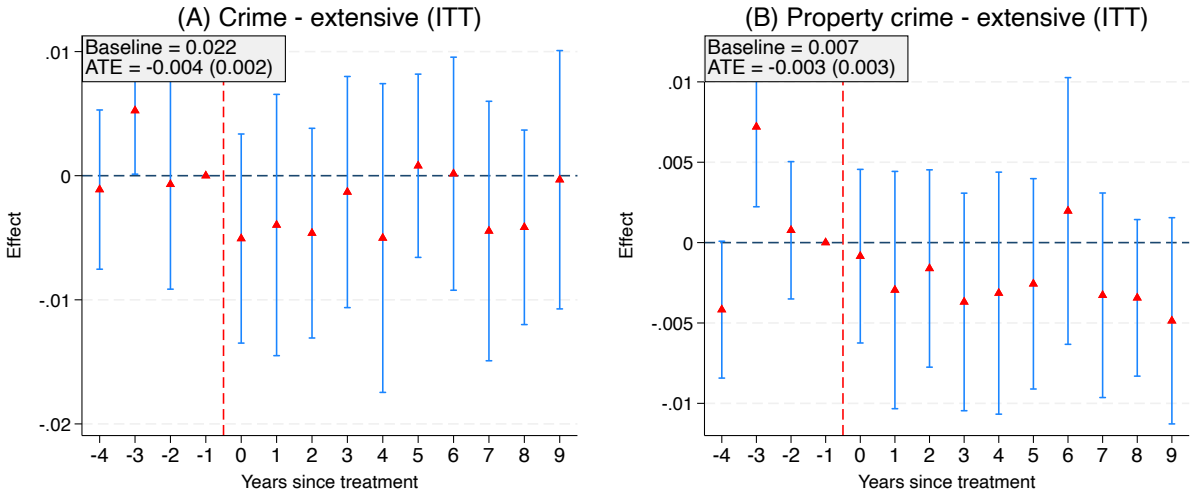
D.2 DDD - WOZ sample

Figure A18: **Event study DDD on incumbents' labour market outcomes - extensive margin**



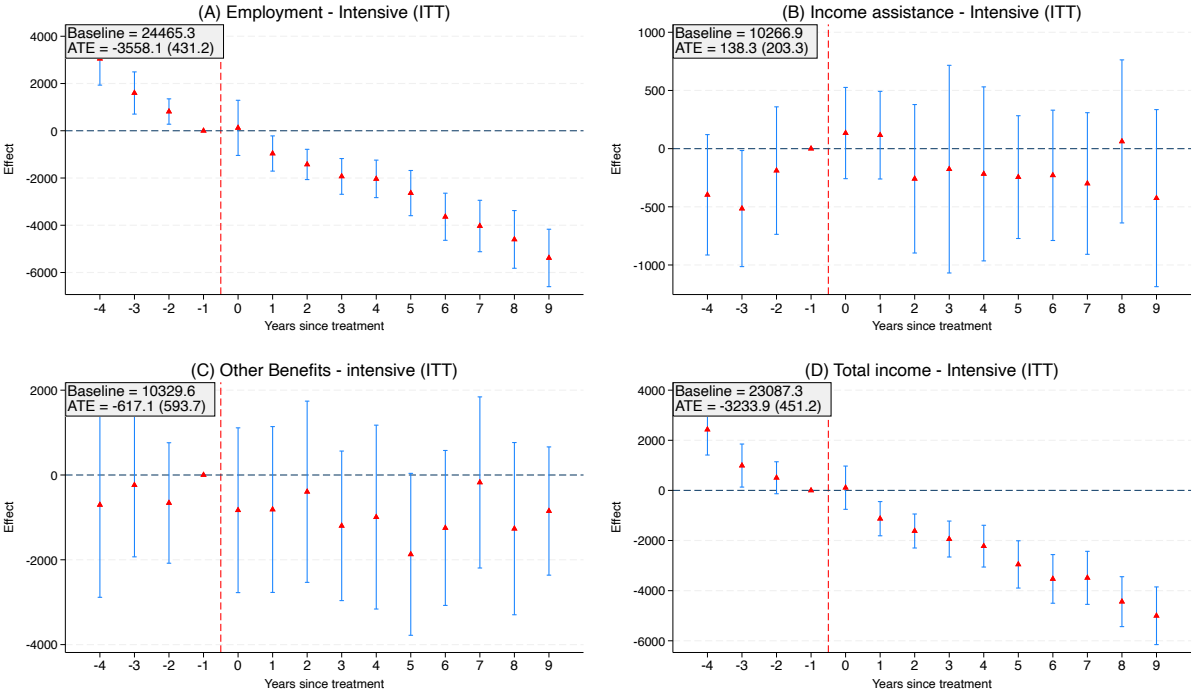
Note: The plots show the results of estimating Equation (1). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on total income at the intensive margin (including zeroes). The baselines are the 2005 variable mean for incumbents in *treated* neighbourhoods.

Figure A19: **Event study DiD on crime for incumbents**



Note: The plots show the results of estimating Equation (2). Panel A: effects on probability to commit a crime. The baseline is the 2005 variable mean for incumbents in *treated* neighbourhoods.

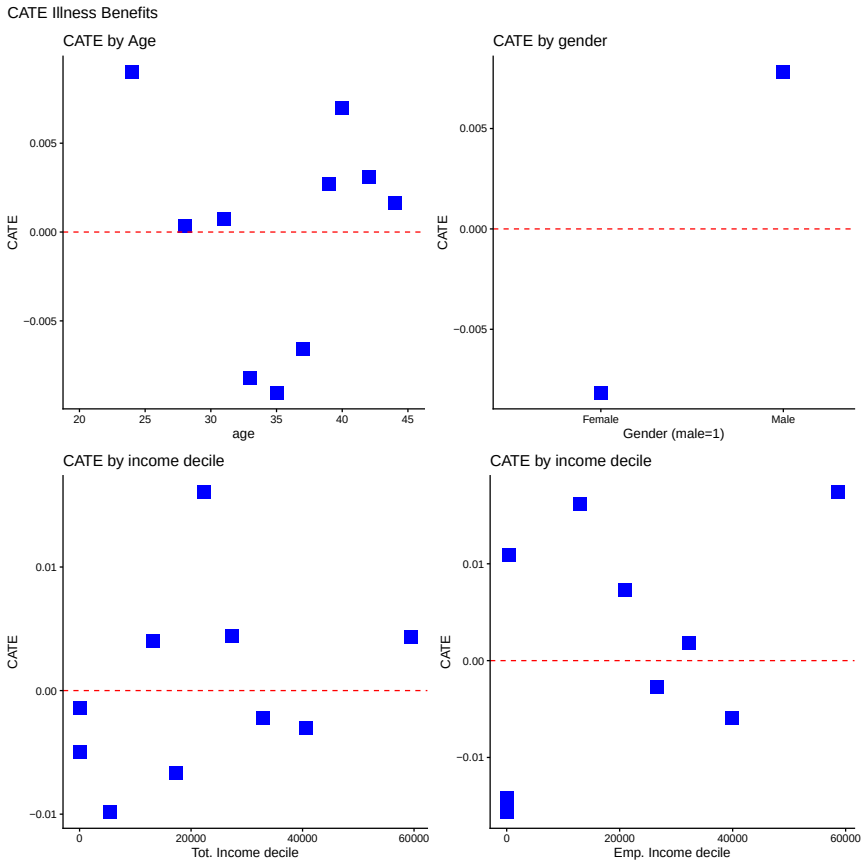
Figure A20: **Event study DDD on incumbents' labour market outcomes - extensive margin**



Note: The plots show the results of estimating Equation (2). Panel A: effects on employment at the extensive margin. Panel B: effects on welfare take-up at the extensive margin. Panel C: effects on other benefit take up (DI and UI) at the extensive margin. Panel D : effects on total income at the intensive margin (including zeroes). The baselines are the 2005 variable mean for incumbents in *treated* neighbourhoods.

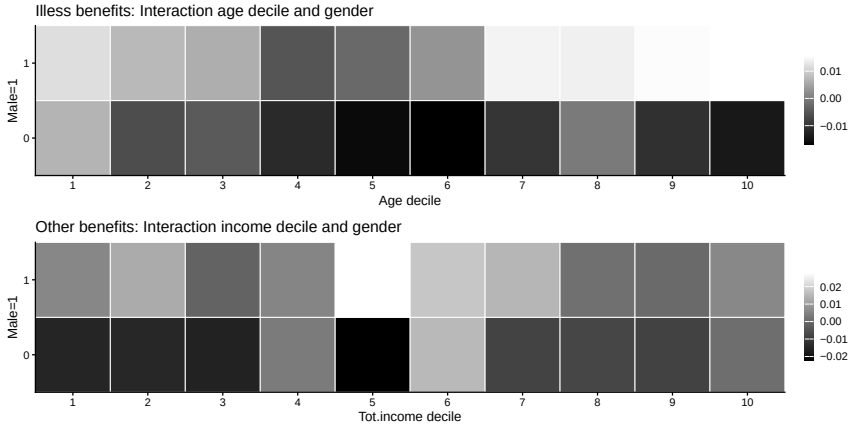
E Heterogeneity - DDD

Figure A21: **CATE on other benefits - extensive margin**



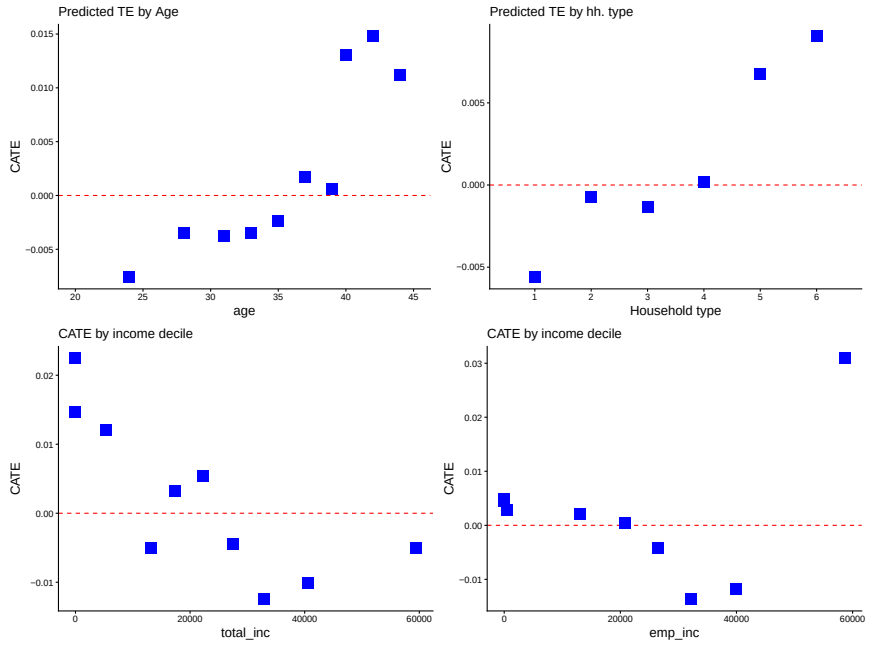
Note: CATE on welfare participation by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

Figure A22: **CATE on other benefits by pairs of characteristics - extensive margin**



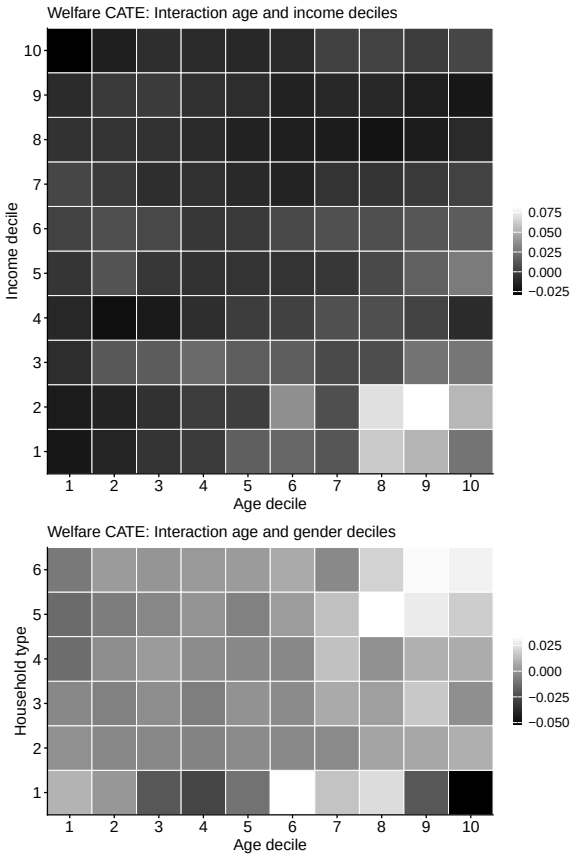
Note: CATE on welfare participation by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

Figure A23: **CATE on welfare participation by characteristic - extensive margin**



Note: Result of estimating 2x2 DiD setting on incumbents sample. Panel A: mean CATE by age decile. Panel B: mean CATE by gender. Panel C: mean CATE by migration generation. Panel D: mean CATE by household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

Figure A24: **CATE on welfare participation by pairs of characteristics - extensive margin**

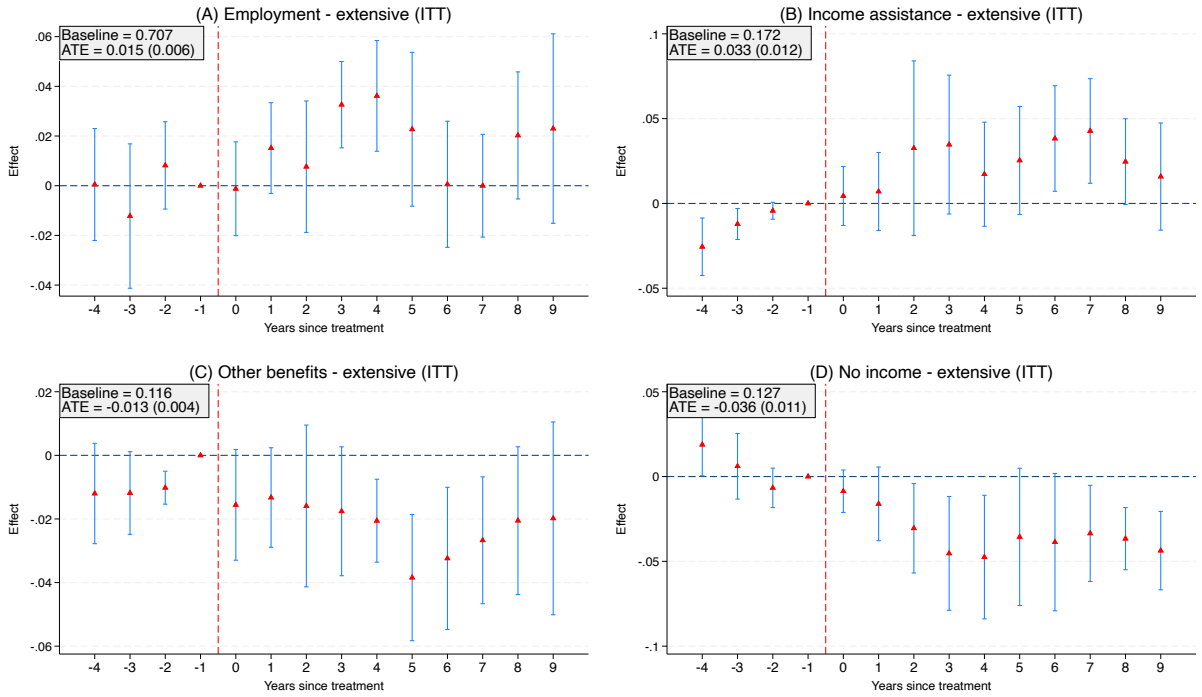


Note: CATE on welfare participation by pairs of characteristics. Age is binned by decile. The top panel shows the interaction between age decile and gender. The bottom panel shows the interactions between age decile and household type. Household types: (1) single person household; (2) non-married couple, no kids (3) married couple, no children; (4) non-married with kids; (5) married with kids; (6) single parent with children.

F Robustness checks & spillovers

F.1 Indirect spillovers

Figure A25: **Event study DiD on labour market outcomes for incumbents in spillover neighbourhoods**

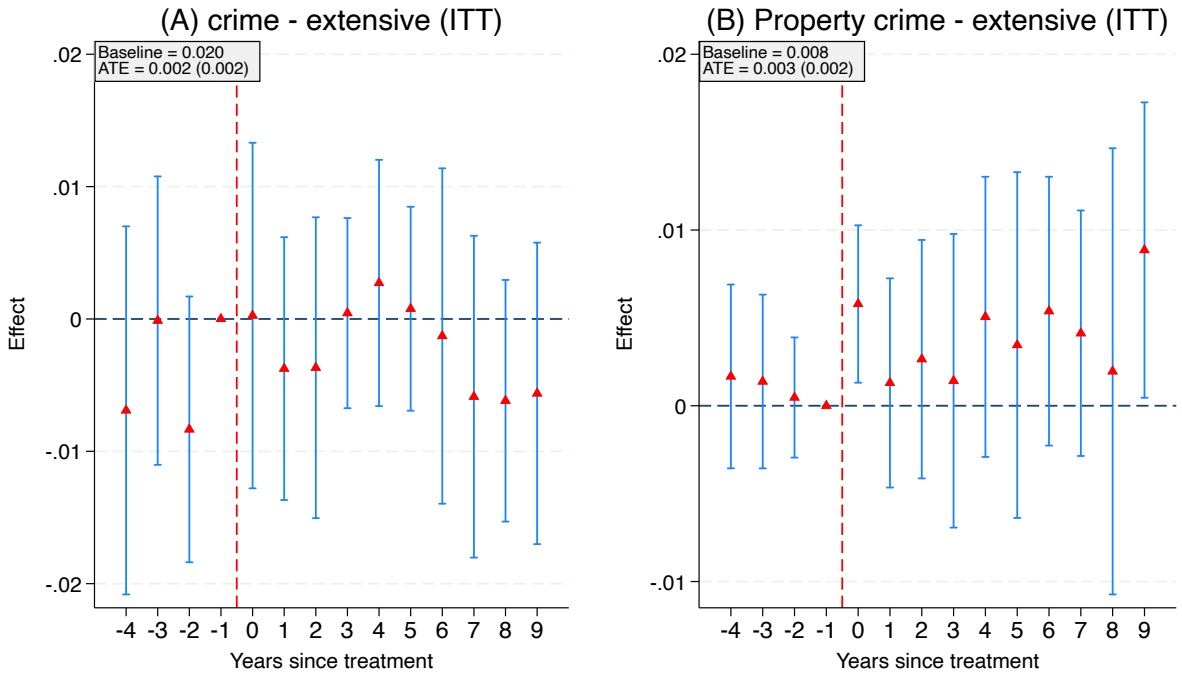


Note: The plots show the results of estimating equation 2 on employment (panel A), welfare benefits (panel B), other benefits (panel C), and total income (panel D) at the extensive margin. All adult individuals living in spillover neighbourhoods are treated. All adults living in remaining untreated neighbourhoods (non-treated) are the controls.

The results of the second robustness check suggest two possible interpretations. Firstly, our original estimates could be capturing a general shock in neighbourhoods south of Rotterdam that are not absorbed by individual, neighbourhood, or time fixed effects. We could not find a shock capable of explaining this result.[19] A second more plausible explanation is related to the fact that spillover neighbourhoods directly surround treated neighbourhoods (see Map A29). The effects of the *Rotterdam Wet* likely spilled over to neighbouring areas, such that a treatment effect is detectable for residents in adjacent neighbourhoods. The lower magnitude of the results in Figure **??** is consistent with a treatment effect decaying farther away from treated areas. If this hypothesis is correct, our main specification underestimates the treatment effect. Future steps in this research should establish whether and how the effects of the policy spilled over onto adjacent areas.

[19] Another place-based policy (the *Nationaal Programma Rotterdam Zuid* was implemented in some of the treated neighbourhoods from 2013-2014 onwards, and should not affect our results in the first years post-treatment.

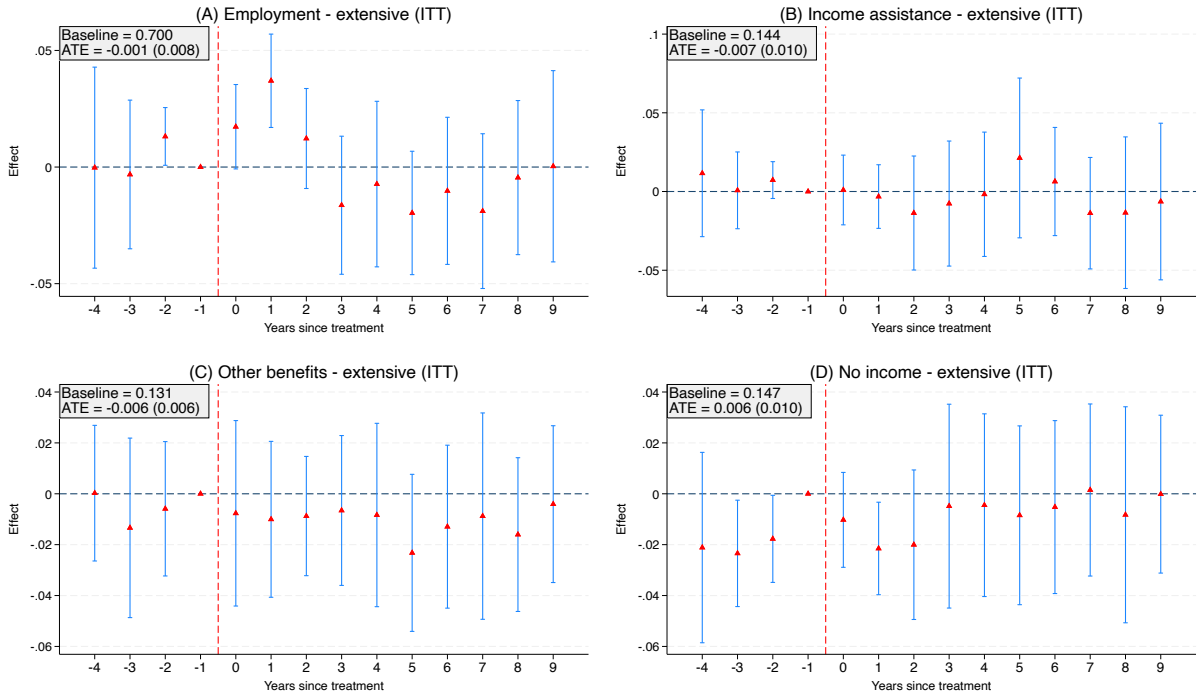
Figure A26: **Event study DiD on labour market outcomes for incumbents in spillover neighbourhoods**



Note: The plots show the results of estimating equation 2 on employment (panel A), welfare benefits (panel B), other benefits (panel C), and total income (panel D) at the extensive margin. All adult individuals living in spillover neighbourhoods are treated. All adults living in remaining untreated neighbourhoods (non-treated) are the controls.

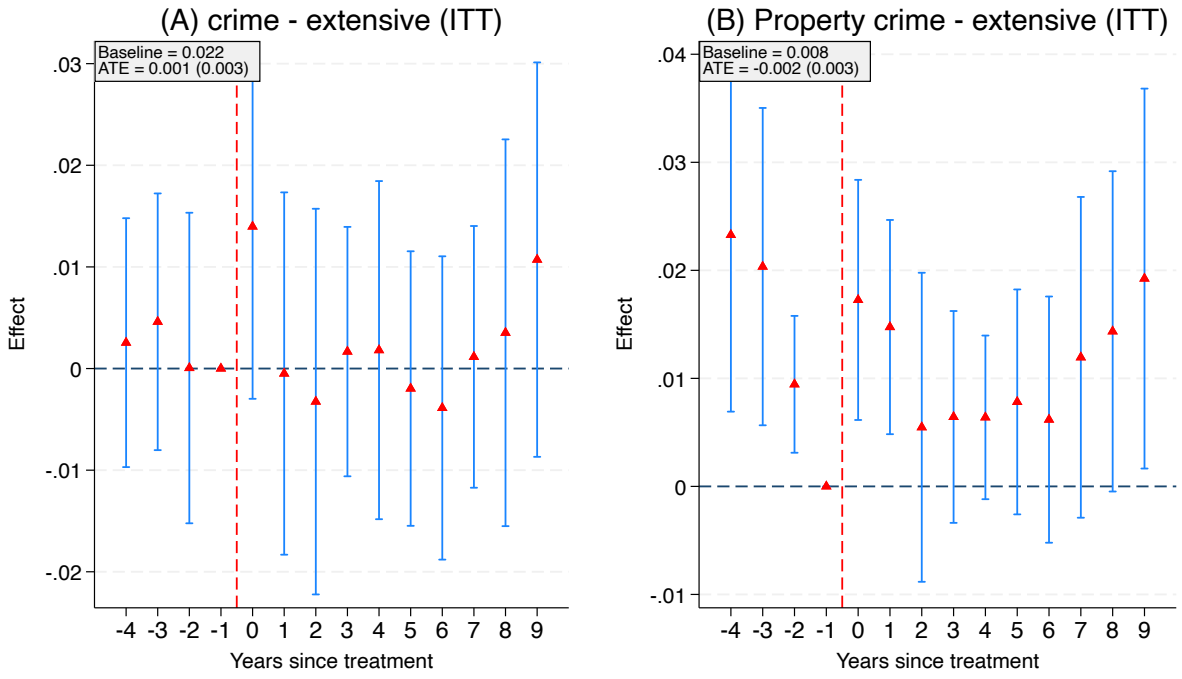
F.2 Placebo

Figure A27: **Event study DiD on labour market outcomes for incumbents in spillover neighbourhoods**



Note: The plots show the results of estimating equation 2 on employment (panel A), welfare benefits (panel B), other benefits (panel C), and total income (panel D) at the extensive margin. All adult individuals living in spillover neighbourhoods are treated. All adults living in remaining untreated neighbourhoods (non-treated) are the controls.

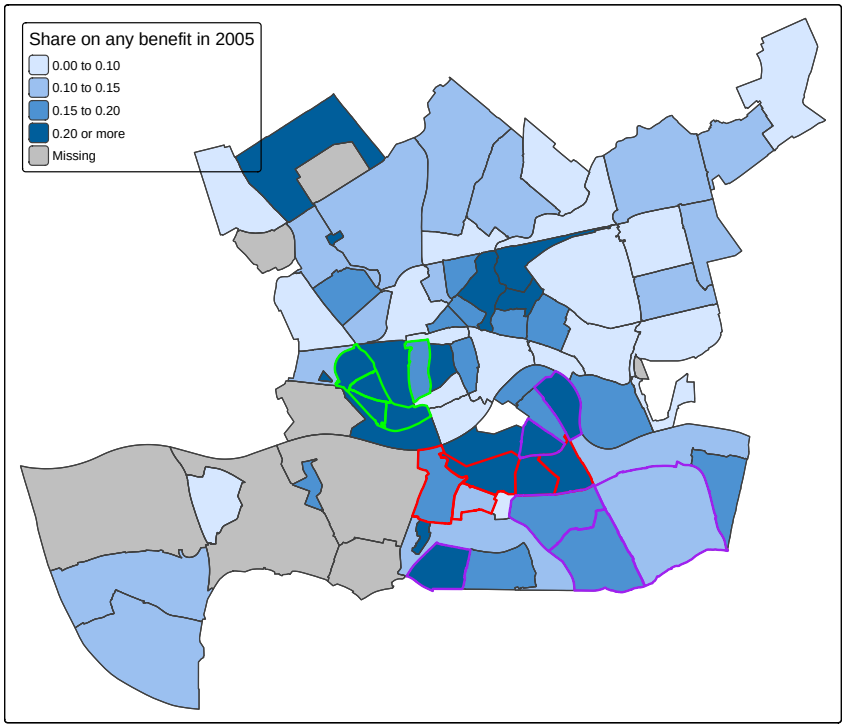
Figure A28: **Event study DiD on labour market outcomes for incumbents in spillover neighbourhoods**



Note: The plots show the results of estimating equation 2 on employment (panel A), welfare benefits (panel B), other benefits (panel C), and total income (panel D) at the extensive margin. All adult individuals living in spillover neighbourhoods are treated. All adults living in remaining untreated neighbourhoods (non-treated) are the controls.

F.3 Map of treated and spillover areas

Figure A29: **Map of fully treated and spillover neighbourhoods**



Fully treated (red borders), partially treated (green borders), and 'spillover' neighbourhoods (purple borders). The filling shows the share of individuals on any benefit in Rotterdam by treatment status.

G Additional information on the Dutch welfare system

Unemployment Unemployment insurance (henceforth UI) is funded through payroll contributions andad–ministered by the employee insurance agency, the UWV (Overheid, 2025a). To claim UI, an individual must be involuntarily unemployed and have worked 26 out of the last 36 weeks before unemployment. The benefits are earnings-related. In the first two months of unemployment, recipients receive 75% of their wage and 70% thereafter. The entitlement period depends on the previous job tenure, and varies between 3 months and 2 years. Legally employed migrants have the same rights as nationals.

Disability and Sickness Disability insurance, governed by the WIA (Overheid, 2025c), is aimed at workers unable to work after a long-term illness. After two years of illness, an employee able to earn 65% or less of their previous wage may receive support. The insurance system guarantees 70% of the previous wage, although compensations vary by illness. Nationals and legal migrant workers have equal rights to disability insurance (henceforth DI). Sickness benefits can be claimed if an employers has no legal obligation to pay wages after an employee is declared as sick.

Welfare Welfare (bijstand) is administered by municipalities. It provides a means-tested allowance for individuals ineligible to other benefits. Recipients must be between 18 and the pension age. The allowance is 70% of the minimum wage for an adult living alone, and varies with the number of dependent children. Foreigners qualify to welfare benefits if they have lived in the Netherlands for more than 3 months, although

the immigration authority may revoke residence permits for inactive foreigners. The welfare benefit system was reformed in 2005 (Overheid, 2025b). The reform transferred budget competences to municipalities, but did not fundamentally alter eligibility criteria.